Check for updates

Received: 21 October 2022 | Accepted: 17 October 2023

DOI: 10.1111/bmsp.12327

the british psychological society
promoting excellence in psychology

ARTICLE

The effective sample size in Bayesian information criterion for level-specific fixed and random-effect selection in a two-level nested model

Sun-Joo Cho[1] | **Hao Wu**[1] | **Matthew Naveiras**[2]

[1]Vanderbilt University, Nashville, Tennessee, USA

[2]Riverside Insights, Itasca, Illinois, USA

Correspondence
Sun-Joo Cho, Vanderbilt University, Nashville, TN, USA.
Email: sj.cho@vanderbilt.edu

Abstract

Popular statistical software provides the Bayesian information criterion (BIC) for multi-level models or linear mixed models. However, it has been observed that the combination of statistical literature and software documentation has led to discrepancies in the formulas of the BIC and uncertainties as to the proper use of the BIC in selecting a multi-level model with respect to level-specific fixed and random effects. These discrepancies and uncertainties result from different specifications of sample size in the BIC's penalty term for multi-level models. In this study, we derive the BIC's penalty term for level-specific fixed- and random-effect selection in a two-level nested design. In this new version of BIC, called BIC_{E1}, this penalty term is decomposed into two parts if the random-effect variance–covariance matrix has full rank: (a) a term with the log of average sample size per cluster and (b) the total number of parameters times the log of the total number of clusters. Furthermore, we derive the new version of BIC, called BIC_{E2}, in the presence of redundant random effects. We show that the derived formulae, BIC_{E1} and BIC_{E2}, adhere to empirical values via numerical demonstration and that BIC_E (E indicating either $E1$ or $E2$) is the best global selection criterion, as it performs at least as well as BIC with the total sample size and BIC with the number of clusters across various multi-level conditions through a simulation study. In addition, the use of BIC_{E1} is illustrated with a textbook example dataset.

KEYWORDS

Bayesian information criterion, level-specific fixed effects, linear mixed models, mixed model, model selection, multi-level model, random effect

Sun-Joo Cho and Hao Wu contributed equally to this work.

 © 2023 British Psychological Society.

1 | INTRODUCTION

1.1 | Bayesian information criterion for multi-level models

Multi-level models (MLMs; e.g., Goldstein, 2003) or linear mixed models (Laird & Ware, 1982) are a general class of modelling framework used to describe the relationship between the response variable and covariates for clustered data that includes repeated measures and nested designs. Selecting fixed and random effects is an important step in the application of MLMs. When competing MLMs are compared in model selection, information criteria such as the Bayesian information criterion (BIC; Schwarz, 1978) have been used for MLMs (e.g., Hamaker et al., 2011; McCoach & Black, 2008; Whittaker & Furlow, 2009). Most software for MLMs, including SPSS (IBM Corp., 2020), the MIXED procedure in SAS (SAS Institute Inc., 2015), the R library `lme4` (Bates et al., 2015), and Mplus (Muthén & Muthén, 1998–2015), provide BIC and/or its sample-size modification as a part of its output.

The BIC allows for comparisons among two or more competing models, whether or not they are nested, and it quantifies the degree to which a given model represents an improvement over the other competing models (Burnham & Anderson, 2002). The BIC is a consistent criterion (e.g., see Claeskens & Hjort, 2008, for a discussion on consistency vs. efficiency for information criteria), meaning that the BIC will asymptotically select the true model if considered. The BIC is relatively easy to calculate, and under certain conditions the difference between two BICs is a rough approximation to the logarithm of the Bayes factor whose computation would require prior distributions (Berger et al., 2003; Kass & Raftery, 1995). A comparison using the BIC does not constitute a statistical test of the difference in the competing models, and model selection is done based on a ranking of the BIC values among competing models.

However, it is challenging to use the BIC for MLMs because the sample size in the BIC's penalty term is not clearly defined. For an MLM, the sample size in BIC calculation has been inconsistently defined in the literature to be the total sample size, the number of clusters, the cluster sizes, or the weighted average of the total sample size and cluster sizes (e.g., Hamaker et al., 2011, p. 249; McCoach & Black, 2008, p. 253). The sample size in the BIC's penalty term has similarly been inconsistently defined across software: SPSS, R, and Mplus use the total sample size for all kinds of MLMs, whereas the MIXED procedure in SAS (SAS Institute Inc., 2015, p. 6064) uses the number of levels of the first random effect specified in a RANDOM statement for the sample size (e.g., the number of clusters for a two-level random intercept model). Furthermore, MLM textbooks also differ in their recommendations. For example, Hox et al. (2018, p. 39) and Goldstein (2003, p. 37) noted that the total number of higher-level units in an MLM is often used as an approximation of the effective sample size. However, Snijders and Bosker (2012, p. 202) presented BIC with the total sample size in MLM applications. When the sample size in the BIC's penalty term is not correctly defined for MLMs, different calculations of BIC result in different model selection results.

1.2 | Limitations of existing BICs for MLMs and related models

For MLMs, observations within a cluster tend to be dependent. This affects model selection with BIC because the effective total sample size tends to be smaller than the total number of observations. Several researchers have attempted to calculate an *effective sample size* for dependent observations. Pauler (1998) studied the effective sample size when comparing normal linear mixed models with different fixed effects and showed that for fixed effects with no associated random effects, the effective sample size is the total sample size, but for fixed effects with associated random effects it is the number of clusters. However, the author did not consider the effective sample size for the covariance parameters of random effects. Kass and Raftery (1995, p. 779) and Berger et al. (2003, p. 243) noted that the effective sample size could be derived as the scalar for an approximation

20448317, 2024, 2, Downloaded from https://bpspsychub.onlinelibrary.wiley.com/doi/10.1111/bmsp.12327 by University Of North Florida, Wiley Online Library on [22/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

20448317, 2024, 2, Downloaded from https://bpspsychub.onlinelibrary.wiley.com/doi/10.1111/bmsp.12327 by University Of North Florida, Wiley Online Library on [22/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

of the information matrix. However, these previous studies cannot be applied directly to MLMs. Jones (2011) proposed using the total elements in the inverse of each cluster's correlation matrix (of the responses) summed across clusters (equation 3 in Jones, 2011) as the effective sample size in the BIC's penalty term for a linear mixed model. Despite the fact that the author showed that the effective sample size was a function of the intraclass correlation (ICC) (equation 6 in Jones, 2011) for a random intercept model, the author did not show the effective sample size in the context of selecting the level-specific fixed and random effects (random intercept and random slope) of the linear mixed model. Delattre et al. (2014) derived a BIC for linear and non-linear mixed models. Their derived BIC involves both the fixed-effect parameters and the (co)variance parameters in the random-effect covariance matrix and is consistent with the findings of Pauler (1998) for the fixed effects in a linear mixed model. However, their derivation is based on a specific form of a mixed model rather than a general representation of a model with separate design matrices for fixed and random effects. Lorah and Womack (2019) showed via a simulation study that the BIC produces appropriate model selection behaviour regarding *fixed* effects in the case where the effective sample size is the number of clusters for between-cluster fixed effects and is the total sample size for within-cluster fixed effects in a two-level random intercept model. Although this sheds light on the importance of using the appropriate effective sample size for the level-specific fixed effects, its results are not extended to more general situations (e.g., models with random slopes). In an unpublished paper, Shen and González (2021) advocated for the use of the sum of all entries of the inverse correlation matrix of the parameter estimates as the effective sample size. However, a proof of this was not provided. To summarize, there is no prior research analytically deriving the effective sample size in the BIC for the most general form of linear mixed models in the *joint* selection of level-specific fixed and random effects (random intercepts and random slopes) with covariances. The joint selection allows level-specific fixed effects to be selected when the structure of the random effects is unknown.

1.3 | Purpose and novel contributions of the current study

The purpose of this paper is to derive the BIC's penalty term for an MLM in its most general form to facilitate the joint selection of level-specific fixed and random effects. Specifically, we analyse the large sample behaviour of the information matrix in the MLM for two-level nested data (e.g., students nested within schools).

The contribution of the current study is fourfold. First, we derive the BIC's penalty term for both the fixed-effect parameters and the (co)variance parameters, assuming the variance–covariance matrix of random effects has full rank. The BIC based on this derived penalty term, called BIC_{E1}, can be used to select a model regarding level-specific fixed effects and random effects (random intercept, random slopes, and their covariances) as a general model using the 'correct' BIC in applications. Second, a new version of BIC, called BIC_{E2}, is provided that is recommended in the presence of redundant random effects. Redundant random effects can result when researchers fit the most complex MLMs possible consistent with their study designs, such as including the random slopes of covariates related to all experimental conditions following the 'keep in maximal' recommendation by Barr et al. (2013). Third, this study demonstrates that theoretical predictions of BIC_{E1} and BIC_{E2} adhere to empirical values in various conditions (which will be identified based on their derivations) that may affect the behaviour of BIC_{E1} and BIC_{E2}. Fourth, the relative performance of the modified BIC (BIC_{E1} and BIC_{E2}) for MLMs is evaluated by comparing existing BICs empirically via a simulation study.

The remainder of this paper is organized as follows. In Section 2, the BIC and Laplace approximation as its theoretical basis are reviewed. In Section 3, we derive the BIC_{E1} for a two-level nested model when the variance–covariance matrix of random effects has full rank. In Section 4, we derive the BIC_{E2} for models with redundant random effects. In Section 5, the finite sample performance of BIC_{E1} and

BIC_{E2} is evaluated. In Section 6, the relative performances of the BIC_{E1} and BIC_{E2} are compared with those of existing BICs in various multi-level designs. In Section 7, the use of the BIC_{E1} is illustrated using a textbook example. In Section 8, we end with a summary and a discussion.

2 | BAYESIAN INFORMATION CRITERION

Schwarz (1978) derived the BIC as an asymptotic approximation of the Bayesian marginal probability of a candidate model M. The original derivation in Schwarz (1978) was 'for the case of independent, identically distributed (i.i.d.) observations, and linear models', under the assumption that the likelihood belongs to an exponential family. It has been noted that BIC can be used for observations not necessarily identically distributed (e.g., Pauler, 1998; Stone, 1979). In what follows, the notation for the candidate model M is omitted for simplicity.

The BIC can be derived based on Laplace approximation (e.g., Kass & Vaidyanathan, 1992; Raftery, 1995), which is also valid for mixed models (e.g., Wolfinger, 1993). In the Laplace approximation method, the log likelihood, $g(\boldsymbol{\vartheta}) = \log f(\mathbf{y}|\boldsymbol{\vartheta})$, is expanded as a quadratic Taylor series about the maximum likelihood estimate (MLE) $\widehat{\boldsymbol{\vartheta}}$ under regularity conditions (de Bruijn, 1970, sect. 4.4; Tierney & Kadane, 1986). Taking the logarithm of the marginal probability $f(\mathbf{y})$ with the Laplace approximation leads to

$$\log f(\mathbf{y}) = \log f(\mathbf{y}|\widehat{\boldsymbol{\vartheta}}) + \log\pi(\widehat{\boldsymbol{\vartheta}}) + \frac{K}{2}\log(2\pi) - \frac{1}{2}\log|\mathbf{A}| + o_p(1), \tag{1}$$

where $\mathbf{y}$ is data, $f(\mathbf{y}|\widehat{\boldsymbol{\vartheta}})$ is the likelihood of data $\mathbf{y}$ evaluated at $\widehat{\boldsymbol{\vartheta}}$, $\pi(\widehat{\boldsymbol{\vartheta}})$ is the prior density, K is the number of parameters, $\mathbf{A}$ is the information matrix, and $o_p(1)$ denotes a term that converges to zero in probability as $n \to \infty$. We note that the second and third terms are of order $O_p(1)$. When observations are i.i.d., $\log|\mathbf{A}| = K\log(N) + O_p(1)$, where N is the sample size. For a large sample size, the BIC can be presented as

$$\mathrm{BIC} = -2\log f(\mathbf{y}|\widehat{\boldsymbol{\vartheta}}) + K\log N. \tag{2}$$

3 | BIC FOR A TWO-LEVEL NESTED MODEL

For two-level nested data, the response vector $\mathbf{y}_j$ is composed of the responses y_{ij} for observation i ($i = 1, \ldots, n_j$) in cluster j ($j = 1, \cdots, J$), where n_j is the sample size for cluster j. The standard form of the variance component model (Laird & Ware, 1982) is

$$\mathbf{y}_j = \mathbf{X}_j\boldsymbol{\beta} + \mathbf{Z}_j\mathbf{b}_j + \boldsymbol{\epsilon}_j, \tag{3}$$

where $\mathbf{X}_j$ is an $(n_j \times p)$ design matrix for p fixed effects, $\mathbf{Z}_j$ is an $(n_j \times q)$ design matrix for q random effects, $\boldsymbol{\beta}$ is a $p \times 1$ vector of fixed effects, $\mathbf{b}_j$ is a $q \times 1$ vector of random effects distributed as $\mathbf{b}_j \sim \mathrm{MVN}(\mathbf{0}, \boldsymbol{\Psi}_j)$ (where $\boldsymbol{\Psi}_j$ is a variance–covariance matrix of random effects), and $\boldsymbol{\epsilon}_j$ is a $n_j \times 1$ vector of residuals distributed as $\boldsymbol{\epsilon}_j \sim \mathrm{N}(0, \sigma_j^2 I_{n_j})$, where I_{n_j} is an identity matrix of size n_j. In most applications of MLMs, it is often assumed that all $\boldsymbol{\Psi}_j$ and σ_j^2 are constant across all clusters (e.g., de Leeuw & Meijer, 2007). Thus, in this study, we assume that $\mathbf{b}_j \sim \mathrm{MVN}(\mathbf{0}, \boldsymbol{\Psi})$ and $\boldsymbol{\epsilon}_j \sim \mathrm{N}(0, \sigma^2 I_{n_j})$ (homoscedasticity). The population value of $\boldsymbol{\Psi}$ is assumed to have full rank in this section. The situation where $\boldsymbol{\Psi}$ is rank deficient will be discussed later.

The total sample size is denoted by N, calculated as $N = nJ$ for a balanced design ($n_j = n$) and $N = \sum_{j=1}^{J} n_j$ for an unbalanced design. The average cluster size is denoted by $\bar{n} = N/J$. We also define

20448317, 2024, 2, Downloaded from https://bpspsychub.onlinelibrary.wiley.com/doi/10.1111/bmsp.12327 by University Of North Florida, Wiley Online Library on [22/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

20448317, 2024, 2, Downloaded from https://bpspsychub.onlinelibrary.wiley.com/doi/10.1111/bmsp.12327 by University Of North Florida, Wiley Online Library on [22/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

$\boldsymbol{\vartheta} = [\boldsymbol{\beta}', \boldsymbol{\psi}', \sigma^2]'$ as the vector of parameters, where $\boldsymbol{\psi} = \text{vech}(\boldsymbol{\Psi})$ is a vector of the non-duplicated elements in $\boldsymbol{\Psi}$. The total number of parameters is denoted by $K = p + q^* + 1$, where p is the number of fixed-effect parameters in $\boldsymbol{\beta}$ and $q^* = q(q+1)/2$ is the total number of variance and covariance parameters in $\boldsymbol{\Psi}$. In addition, we define the $(q^*+1)\times 1$ vector $\boldsymbol{\rho} = [\text{vech}'(\boldsymbol{\Psi}), \sigma^2]'$, which only contains the random-effect parameters.

For J independent clusters, the matrix $\mathbf{A}$ in Equation (1) can be expressed as

$$\mathbf{A} = \sum_j \mathbf{I}_j(\widehat{\boldsymbol{\vartheta}}),$$

where $\mathbf{I}_j(\widehat{\boldsymbol{\vartheta}})$ is the Fisher information for cluster j. For the remainder of this section, we will focus on the expression of $\mathbf{I}_j$ and the asymptotic order of its elements.

The mean vector $\boldsymbol{\mu}_j$ and covariance matrix $\mathbf{V}_j$ for the marginal distribution of $\mathbf{y}_j$ are given by

$$\boldsymbol{\mu}_j = \mathbf{X}_j\boldsymbol{\beta} \quad \text{and} \quad \mathbf{V}_j = \mathbf{Z}_j\boldsymbol{\Psi}\mathbf{Z}_j' + \sigma^2 I_{n_j}.$$

When no parameter appears in both the mean and covariance structures, the information matrix of a multivariate normal model is block diagonal, with a block $\mathbf{I}_{\boldsymbol{\beta}\boldsymbol{\beta}}$ for the mean structure parameters (here the fixed effects $\boldsymbol{\beta}$) and a second block $\mathbf{I}_{\boldsymbol{\rho}\boldsymbol{\rho}}$ for the covariance structure parameters (here the residual variance σ^2 and the non-duplicate elements of $\text{vech}'(\boldsymbol{\Psi})$). The log-determinant of the information matrix can be expressed as

$$\log|\mathbf{A}| = \log|\mathbf{I}_{\boldsymbol{\beta}\boldsymbol{\beta}}| + \log|\mathbf{I}_{\boldsymbol{\rho}\boldsymbol{\rho}}|. \tag{4}$$

Below we investigate the two blocks separately. To approximate the first term $\log|\mathbf{I}_{\boldsymbol{\beta}\boldsymbol{\beta}}|$, we decompose the design matrix $\mathbf{X}_j$ into two terms, one in the column space of $\mathbf{Z}_j$ and the other orthogonal to it. We write

$$\mathbf{X}_j = \mathbf{Z}_j\mathbf{W}_j + \mathbf{E}_j$$

for some $q\times p$ matrix $\mathbf{W}_j$ and some $n_j \times p$ matrix $\mathbf{E}_j$ with $\mathbf{E}_j'\mathbf{Z}_j = \mathbf{O}$ (a $p\times q$ zero matrix). From the residual matrix $\mathbf{E}_j$ we define $\mathbf{S}_{EE,n_j} = \mathbf{E}_j'\mathbf{E}_j/n_j$. Assume that $\overline{\mathbf{S}}_{EE} = \frac{1}{N}\sum_j n_j \mathbf{S}_{EE,n_j}$ is bounded and has (limiting) rank p_1, and let $p_2 = p - p_1$. We have the following result:

$$\log|\mathbf{I}_{\boldsymbol{\beta}\boldsymbol{\beta}}| = p_2\log(J) + p_1\log(N) + O_p(1) = p\log(J) + p_1\log(\overline{n}) + O_p(1), \tag{5}$$

where $\overline{n}$ is the average cluster size. Details of the derivation can be found in Appendix A.1.

It is important to understand the rank of $\overline{\mathbf{S}}_{EE}$ and to determine what kinds of fixed effects contribute to it. Let $\mathbf{E} = [\mathbf{E}_1', \mathbf{E}_2', \ldots, \mathbf{E}_J']'$ be the $N\times p$ matrix formed by stacking the individual $\mathbf{E}_j$. The rank of $\overline{\mathbf{S}}_{EE}$ is the column rank of this matrix $\mathbf{E}$. The matrix $\mathbf{E}$ can also be considered as the residuals of the $N\times p$ matrix $[\mathbf{X}_1', \mathbf{X}_2', \ldots, \mathbf{X}_J']'$ after regressing its columns onto the columns of the block diagonal matrix $\mathbf{diag}\left\{\mathbf{Z}_1, \mathbf{Z}_2, \ldots, \mathbf{Z}_J\right\}$. Intuitively this is the number of fixed effects that have no associated random effect. The following types of fixed effects are considered associated with a random effect contributing to p but do *not* contribute to p_1:

- The intercept (represented by a column of 1's): It has both a fixed and random effect and is present in both $\mathbf{X}_j$ and $\mathbf{Z}_j$.

- Between-cluster covariates: These correspond to columns of constants within $\mathbf{X}_j$.
- Within-cluster covariates with both fixed and random effects: They specify identical columns in $\mathbf{X}_j$ and $\mathbf{Z}_j$.
- Interaction effects between between-cluster covariates and within-cluster covariates with random effects: Each of them corresponds to a column of $\mathbf{X}_j$ that is a scalar multiple of a column of $\mathbf{Z}_j$.

These columns of $\mathbf{X}_j$ are entirely within the column space of $\mathbf{Z}_j$, so their corresponding columns in $\mathbf{E}_j$ are zeros. In contrast, the following types of fixed effects *do* contribute to p_1:

- Within-cluster covariates with fixed effects (but not random effects): These correspond to nonzero columns within $\mathbf{E}_j$, contributing to the rank of $\mathbf{S}_{EE}$.
- An interaction between a between-cluster covariate and a within-cluster covariate that has a fixed effect but no random effect: Such an interaction produces a non-zero column of $\mathbf{E}_j$ that is a scalar multiple of another column of $\mathbf{E}_j$. While they do not contribute to the column rank of $\mathbf{E}_j$, they do contribute to the column rank of $\mathbf{E}$ and, therefore, to the rank of $\mathbf{S}_{EE}$.

We now turn to the second term in Equation (4), $\log|\mathbf{I}_{\boldsymbol{\rho\rho}}|$. Our derivations in Appendix A.2 lead to

$$\log|\mathbf{I}_{\boldsymbol{\rho\rho}}| = q^*\log(J) + \log(N) + O_p(1) = (q^* + 1)\log(J) + \log(\overline{n}) + O_p(1), \tag{6}$$

where $q^* = q(q+1)/2$ is the number of non-duplicated elements in $\boldsymbol{\Psi}$.

Equations (1), (4), (5), and (6) lead to

$$\log f(\mathbf{y}) = f(\mathbf{y}|\widehat{\boldsymbol{\vartheta}}) - \frac{1}{2}[K_1\log(N) + K_2\log(J)] + O_p(1), \tag{7}$$

where $K_1 = p_1 + 1$ and $K_2 = p_2 + q^*$ are the number of parameters that have effective sample sizes of N and J respectively. It is common to formulate BIC with deviance as follows:

$$\mathrm{BIC}_{E1} = -2\log f(\mathbf{y}|\widehat{\boldsymbol{\vartheta}}) + K_1\log(N) + K_2\log(J) = -2\log f(\mathbf{y}|\widehat{\boldsymbol{\vartheta}}) + K_1\log(\overline{n}) + K\log(J), \tag{8}$$

in which the subscript $E1$ stands for 'effective sample size' when $\boldsymbol{\Psi}$ has full rank, and $K = K_1 + K_2$ is the total number of parameters. The total penalty term in the conventional BIC (Equation 2) is decomposed into two terms in BIC_{E1}, (a) $K_1\log(N)$ and (b) $K_2\log(J)$ or (a) $K_1\log(\overline{n})$ and (b) $K\log(J)$.

We note that the preceding conclusions are generally consistent with (yet broader than) those in Pauler (1998) and Delattre et al. (2014). Pauler's (1998) derivation did not consider the variance and covariance parameters. Additionally, Pauler's (1998) derivation assumed an orthogonal $\mathbf{X}_j$, ignoring the presence of any between-cluster covariates. While Delattre et al. (2014) considered between-cluster covariates and the covariance parameters in $\boldsymbol{\Psi}$, their derivations did not use the most general form of the MLM given by Equation (3) and did not consider the residual variance parameter σ^2.

4 | BIC IN THE PRESENCE OF REDUNDANT RANDOM EFFECTS

The derivations presented earlier in Section 3 assume that $\boldsymbol{\Psi}$ has full rank. When $\boldsymbol{\Psi}$ is rank deficient, at least one random effect can be expressed as a function of the remaining random effects. In this case, the

20448317, 2024, 2, Downloaded from https://bpspsychub.onlinelibrary.wiley.com/doi/10.1111/bmsp.12327 by University Of North Florida, Wiley Online Library on [22/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

20448317, 2024, 2, Downloaded from https://bpspsychub.onlinelibrary.wiley.com/doi/10.1111/bmsp.12327 by University Of North Florida, Wiley Online Library on [22/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

number of parameters corresponding to the two different effective sample sizes may deviate from their prescribed values in Equation (8). In this section, we will investigate such situations.

Empirical evidence of redundant random effects may include small variances of these random effects (close to 0, but not exactly 0; e.g., Baayen et al., 2008, p. 395) and/or high correlations between the random effects (close to 1 or −1, but not exactly 1 or −1; e.g., Kreft & De Leeuw, 2002, pp. 49–50). Such empirical evidence results from a *close*-to-singular variance–covariance matrix of random effects.

While in such cases one may consider the removal of the redundant random effects by expressing them in terms of the remaining random effects, this may not always lead to interpretable solutions. More importantly, the different expression of BIC under a rank-deficient $\boldsymbol{\Psi}$ has implications for situations when $\boldsymbol{\Psi}$ technically has full rank but is close to rank deficient. This will be shown later in the numerical demonstration.

When $\boldsymbol{\Psi}$ is singular with rank $q_1 < q$, there are $q_2 = q - q_1$ redundant random effects. In this case, a linear transformation of the columns of $\mathbf{Z}_j$ and a compatible linear transformation of the random effects $\boldsymbol{b}_j$ can be employed to isolate these redundant random effects, so that after such a transformation q_2 random effects in vector $\boldsymbol{b}_{2j}$ have variances of zero, and the remaining q_1 random effects in vector $\boldsymbol{b}_{1j}$ have a covariance matrix of full rank. Note that such a transformation corresponds to a linear re-parametrization of the elements in $\boldsymbol{\Psi}$, which only alters the constant term in the approximation of the log-information matrix and does not change the BIC. Let $\mathbf{Z}_{1j}$ be the columns of the transformed random-effect design matrix corresponding to the q_1 random effects in $\boldsymbol{b}_{1j}$, and let $\mathbf{Z}_{2j}$ be the columns corresponding to the q_2 redundant random effects in $\boldsymbol{b}_{2j}$.

For the block $\mathbf{I}_{\boldsymbol{\beta\beta}}$ for fixed effects, similar to our derivations for a non-singular $\boldsymbol{\Psi}$, we write

$$\mathbf{X}_j = \mathbf{Z}_{1j}\mathbf{W}_{1j} + \mathbf{E}_{1j}$$

for some $q_1 \times p$ matrix $\mathbf{W}_{1j}$ and some $n_j \times p$ matrix $\mathbf{E}_{1j}$ with $\mathbf{E}'_{1j}\mathbf{Z}_j = \mathbf{O}$ (a $p \times q_1$ zero matrix), and define $\mathbf{S}_{E_1E_1,n_j} = \mathbf{E}'_{1j}\mathbf{E}_{1j}/n_j$ and $\overline{\mathbf{S}}_{E_1E_1} = \frac{1}{N}\sum_j n_j \mathbf{S}_{E_1E_1,n_j}$. Let the limiting rank of $\mathbf{S}_{E_1E_1}$ be $\tilde{p}_1$. We have the approximation

$$\log|\mathbf{I}_{\boldsymbol{\beta\beta}}| = \tilde{p}_2\log(J) + \tilde{p}_1\log(N) + O_p(1) = p\log(J) + \tilde{p}_1\log(\overline{n}) + O_p(1), \tag{9}$$

where $\tilde{p}_2 = p - \tilde{p}_1$. Details of the derivations can be found in Appendix B.1. Compared to the situation of a non-singular $\boldsymbol{\Psi}$, the fixed-effect design matrix $\mathbf{X}_j$ is now projected onto a lower-dimensional random-effect design matrix, leading to $\tilde{p}_1 > p_1$ and $\tilde{p}_2 < p_2$.

For the block $\mathbf{I}_{\boldsymbol{\rho\rho}}$ for the random-effect parameters, when cluster sizes are equal ($n_j = n$), we have

$$\begin{aligned}\log|\mathbf{I}_{\boldsymbol{\rho\rho}}| &= (q_1q_2+1)\log(nJ) + q_2^*\log(n^2J) + q_1^*\log(J) + O_p(1)\\ &= (q^*+1)\log(J) + (2q_2^* + q_1q_2 + 1)\log(n) + O_p(1),\end{aligned}$$

where $q^* = q(q+1)/2$ and $q_r^* = q_r(q_r+1)/2$ for $r = 1, 2$. Details of the derivations can be found in Appendix B.2. Note the dramatic increase in the coefficient for $\log(n)$ compared to Equation (6).

Combining the two blocks of the information matrices, we can define

$$\mathrm{BIC}_{E2} = -2\log f(\mathbf{y}|\hat{\boldsymbol{\vartheta}}) + \overline{K}_1\log(n) + K\log(J), \tag{10}$$

where the subscript $E2$ stands for the 'effective sample size' in the presence of redundant random effects. In this equation, $K = p + q^* + 1$ is the total number of parameters and $\overline{K}_1 = \tilde{p}_1 + 1 + q_1q_2 + 2q_2^*$ is the

effective number of parameters associated with $\log(n)$. Remember that p_1 is the limiting rank of $\overline{\mathbf{S}}_{E_1E_1}$ defined earlier, $q_2^* = q_2(q_2+1)/2$, $q_1 = \text{rank}\boldsymbol{\Psi}$, and $q_2 = q - q_1$.

Although we handled the situation of a singular $\boldsymbol{\Psi}$ above, more difficult situations may arise where $\boldsymbol{\Psi}$ is *close to* singular. In these situations, the large sample behaviour of the information matrix would lie between those discussed in Sections 3 and 4. In particular, the coefficient for $\log(n)$ will be greater than K_1 in Equation (8) but smaller than $\overline{K}_1$ in Equation (10).

5 | NUMERICAL DEMONSTRATION

In this numerical demonstration, we study the finite sample performance of BIC_{E1} and BIC_{E2}.

5.1 | Study design

A two-level random-intercept-slope model (called Model A) with balanced cluster sizes ($n_j = n$) was considered as a data-generating model with the following specification:

$$y_{ij} = \beta_0 + \beta_1 x_{ij} + \beta_2 x_j + b_{0j} + b_{1j}x_{ij} + \epsilon_{ij} \quad (i = 1, 2, \ldots, n;\ j = 1, 2, \ldots, J), \tag{11}$$

where $\boldsymbol{\beta} = (\beta_0, \beta_1, \beta_2)' = (57.98, 1.93, -14.57)'$ gives the population fixed effects and random effects $\boldsymbol{b} = (b_{0j}, b_{1j})' \sim \text{MVN}(\mathbf{0}, \boldsymbol{\Psi})$, with

$$\boldsymbol{\Psi} = \begin{pmatrix} \tau_0^2 & \tau_{10} \\ \tau_{01} & \tau_1^2 \end{pmatrix} = \begin{pmatrix} 40.20 & -28.95 \\ -28.95 & 21.58 \end{pmatrix}$$

and the random residual ϵ_{ij} is assumed to follow $\text{N}(0, \sigma^2)$, with $\sigma^2 = 42.78$. These population values were chosen from estimates of a two-level random-intercept-slope model reported in Kreft and De Leeuw (2002, pp. 49–50). The within-cluster covariate (x_{ij}) was generated with a normal distribution with mean 0 and standard deviation 2.07. The between-cluster covariate (x_j), the random effects (b_{0j} and b_{1j}), and the errors of within-cluster units ($\epsilon_{1j}, \epsilon_{2j}, \cdots, \epsilon_{nj}$) were generated from a joint $(n+3)$-variate normal distribution in such a way that their sample mean vector and the sample covariance matrix (with denominator J) matched their population true values. In this way, the MLEs of parameters will match their population true values and the computer program will yield the information matrix $\mathbf{A}$ evaluated at the same $\widehat{\boldsymbol{\vartheta}}$, making it possible to study how $\log|\mathbf{A}|$ changes with J and n free from sampling variability.

In this study, we explored the finite sample performance of BIC_{E1} and BIC_{E2} for Model A under various correlation magnitudes and residual variance values. We considered six magnitudes of correlation between random intercept and random slope, ranging from -1.0 to 0, to study the performance of the new BIC measure when the random-effect covariance matrix was singular, close to singular and far from singularity. We also considered three magnitudes of σ^2 (residual variance), including $\sigma^2 = 42.78$, $\sigma^2 = 10$, and $\sigma^2 = 1$, to study its impact through the presence of the term $(\sigma^2\mathbf{S}_{ZZ,n_j}^{-1}/n_j + \boldsymbol{\Psi})^{-1}$ in many parts of the derivations. These led to a total of 18 sets of true parameters for investigation. Additionally, we examined Model B, a two-level random-intercept model, with the same magnitudes of σ^2 as in Model A. The cluster sizes n and the number of clusters J were varied, considering values of $n = 10, 20, 40$ and $J = 50, 100, 200, 400$. In total, there were 12 different multi-level designs considered in this analysis, based on previous research on MLMs.

20448317, 2024, 2, Downloaded from https://bpspsychub.onlinelibrary.wiley.com/doi/10.1111/bmsp.12327 by University Of North Florida, Wiley Online Library on [22/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

5.2 | Expected results

According to our derivations in Sections 3 and 4, BIC_{E1} and BIC_{E2} are linear functions of $\log(n)$ and $\log(J)$. To investigate whether the coefficients of $\log(n)$ and $\log(J)$ are consistent with the empirical values obtained in this demonstration, the two terms of Equation (4), $\log|\mathbf{I}_{\boldsymbol{\beta\beta}}|$ and $\log|\mathbf{I}_{\boldsymbol{\rho\rho}}|$, are both regressed on $\log(n)$ and $\log(J)$.

For both Models A and B, we expect that the coefficient of $\log(J)$ will always be 3 (the total number of fixed effects) for $\log|\mathbf{I}_{\boldsymbol{\beta\beta}}|$. The coefficient of $\log(J)$ for $\log|\mathbf{I}_{\boldsymbol{\rho\rho}}|$ should be the total number of parameters in the covariance matrix. This is 4 for Model A and 2 for Model B.

The coefficient of $\log(n)$ is more complicated. The coefficient K_1 of $\log(n)$ was expressed below Equations (7) and (10) in terms of the rank q_1 of $\boldsymbol{\Psi}$ and the limiting column rank p_1 of the $N \times p$ matrix $\mathbf{E} = [\mathbf{E}_1', \mathbf{E}_2', \ldots, \mathbf{E}_J']'$, where $\mathbf{E}_j$ contains the residuals from regressing $\mathbf{X}_j$ onto $\mathbf{Z}_j$ (with redundant random effects removed). In both Models A and B, $\mathbf{X}_j$ is given by

$$\mathbf{X}_j = \begin{pmatrix} 1 & x_{1j} & x_j \\ 1 & x_{2j} & x_j \\ \vdots & \vdots & \vdots \\ 1 & x_{nj} & x_j \end{pmatrix},$$

where the three columns correspond to the intercept, the within-cluster covariate, and the between-cluster covariate respectively.

For Model B, $\mathbf{Z}_j = \mathbf{1}_n$ is a single column of 1's. The first and third columns of $\mathbf{X}_j$ are in its column space and produce columns of zero residuals; the second column produces non-zero residuals. As a result, the coefficient of $\log(n)$ in $\log|\mathbf{I}_{\boldsymbol{\beta\beta}}|$ should be $p_1 = 1$. The coefficient of $\log(n)$ in $\log|\mathbf{I}_{\boldsymbol{\rho\rho}}|$ should also be 1, as shown in Equation (6). In practice, the coefficients of $\log(n)$ are expected to be greater than 1 due to the presence of $(\sigma^2 \mathbf{S}_{ZZ,n_j}^{-1}/n_j + \boldsymbol{\Psi})^{-1}$ in the $O_p(J)$ terms in $\log|\mathbf{I}_{\boldsymbol{\beta\beta}}|$ and $\log|\mathbf{I}_{\boldsymbol{\rho\rho}}|$, but they will be closer to 1 with smaller σ^2.

The second row of Table 1 summarizes which parameters count towards which coefficient in the two blocks of the information matrix of Model B. The forms of the three versions of BIC, BIC_E (E indicating either $E1$ or $E2$), BIC_N, and BIC_J, are also listed. BIC_N and BIC_J, which are commonly used in MLM applications, were calculated as

$$\text{BIC}_N = -2\log f(\mathbf{y}|\widehat{\boldsymbol{\vartheta}}) + K\log(N) \tag{12}$$

and

$$\text{BIC}_J = -2\log f(\mathbf{y}|\widehat{\boldsymbol{\vartheta}}) + K\log(J), \tag{13}$$

where K is the total number of parameters.

Model A has two random effects, and the asymptotic behaviour of the coefficient of $\log(n)$ depends on the rank of the 2×2 matrix $\boldsymbol{\Psi}$. When the correlation between the two random effects is not -1, no random effect is redundant, and all three columns of $\mathbf{X}_j$ are in the column space of $\mathbf{Z}_j$ (which contains the first two columns of $\mathbf{X}_j$), so the coefficient of $\log(n)$ should be $p_1 = 0$ for $\log|\mathbf{I}_{\boldsymbol{\beta\beta}}|$. The coefficient of $\log(n)$ for $\log|\mathbf{I}_{\boldsymbol{\rho\rho}}|$ should be 1, as in Model B. The first row of Table 1 provides a summary for Model A. In practice, the coefficients of $\log(n)$ are expected to be greater than these desired values but the deviations are expected to decrease as σ^2 decreases. In addition, as the random-effect correlation decreases from 0 to -1, $\boldsymbol{\Psi}$ becomes closer to singular and the coefficients of $\log(n)$ are expected to increase and become closer to the value stipulated by $\overline{K}_1$ in Equation (10) for a singular $\boldsymbol{\Psi}$.

20448317, 2024, 2, Downloaded from https://bpspsychub.onlinelibrary.wiley.com/doi/10.1111/bmsp.12327 by University Of North Florida, Wiley Online Library on [22/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

TABLE 1 Numerical demonstration: Summary of data-generating models and calculations of K_1 and K_2 in BIC_E with comparisons to BIC_N and BIC_J.

	Model A	Model B
Fixed effects	x_{ij}, x_j	x_{ij}, x_j
Random effects	Intercept, Slope	Intercept
Fixed effects for K_1	[0]	β_1[1]
Fixed effects for K_2	β, β_1, β_2[3]	β, β_2[2]
Random effects for K_1	σ^2[1]	σ^2[1]
Random effects for K_2	$\tau_0^2, \tau_1^2, \tau_{01}$[3]	τ_0^2[1]
BIC_{E1}	$D + [1\log(N) + 6\log(J)]$	$D + [2\log(N) + 3\log(J)]$
BIC_N	$D + 7\log(N)$	$D + 5\log(N)$
BIC_J	$D + 7\log(J)$	$D + 5\log(J)$

Note: D indicates deviance ($-2\log f(\mathbf{y}|\hat{\boldsymbol{\vartheta}})$); numbers in square brackets indicate K_1 or K_2; $N = nJ$ is the total sample size.

When the random-effect correlation in Model A is -1, we have $b_{0j} = -b_{1j}$. In this case, we can reparametrize the random effects as $\frac{1}{2} \times (b_{0j} - b_{1j}, b_{0j} + b_{1j})'$, and the new design matrix can be written $\mathbf{Z}_j = [\mathbf{Z}_{1,j}, \mathbf{Z}_{2,j}]$, with $\mathbf{Z}_{1,j} = (1 - x_{1j}, 1 - x_{2j}, \ldots, 1 - x_{nj})'$ and $\mathbf{Z}_{2,j} = (1 + x_{1j}, 1 + x_{2j}, \ldots, 1 + x_{nj})'$. Note that $\mathbf{Z}_{2,j}$ now corresponds to a redundant random effect with a variance of zero. $\mathbf{Z}_{1,j}$ is in the column space of $\mathbf{X}_j$, so the coefficient of $\log(n)$ for $\log|\mathbf{I}_{\boldsymbol{\beta}\boldsymbol{\beta}}|$ is now $p_1 = 2$. For $\log|\mathbf{I}_{\boldsymbol{\rho}\boldsymbol{\rho}}|$, because $q_1 = q_2 = 1$ and $q_2^* = q_2(q_2 + 1)/2 = 1$ in this case, the coefficient of $\log(n)$ is $2q_2^* + q_1 q_2 + 1 = 4$.

5.3 | Results

In Figure 1, the regression coefficients of $\log(n)$ in Model A closely matched their expected values for fixed and random effects, as indicated by BIC_{E2} when the random effects are perfectly correlated. Empirically, regression coefficients decreased to their theoretical values in BIC_{E1} as random-effect correlation and σ^2 decreased, as we expected. For both fixed and random effects, the regression coefficients of $\log(J)$ coincided with their desired values. In Model B (not graphed), the regression coefficients of $\log(n)$ for fixed effects closely aligned with the derived value $K_1 = 1$, and $\log(J)$ matched $K_1 + K_2 = 3$. For random effects, $\log(n)$ matched $K_1 = 1$, and $\log(J)$ agreed with the expected value $K_1 + K_2 = 2$. These results demonstrate consistency between derived effective sample size values and empirical results.

6 | SIMULATION STUDY

In this section, the behaviours of BIC_{E1} and BIC_{E2} are examined via a simulation study in selecting level-specific fixed and random effects. Their behaviours are compared with those of BIC_N and BIC_J in Equations (12) and (13) respectively.

6.1 | Simulation design

Two data-generating models were considered. As the first data-generating model, the random-intercept-slope model (Equation 11) was selected. In the random-intercept-slope model, the same true parameters except the correlation between b_0 and b_1 and the residual variance were used as in

20448317, 2024, 2, Downloaded from https://bpspsychub.onlinelibrary.wiley.com/doi/10.1111/bmsp.12327 by University Of North Florida, Wiley Online Library on [22/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

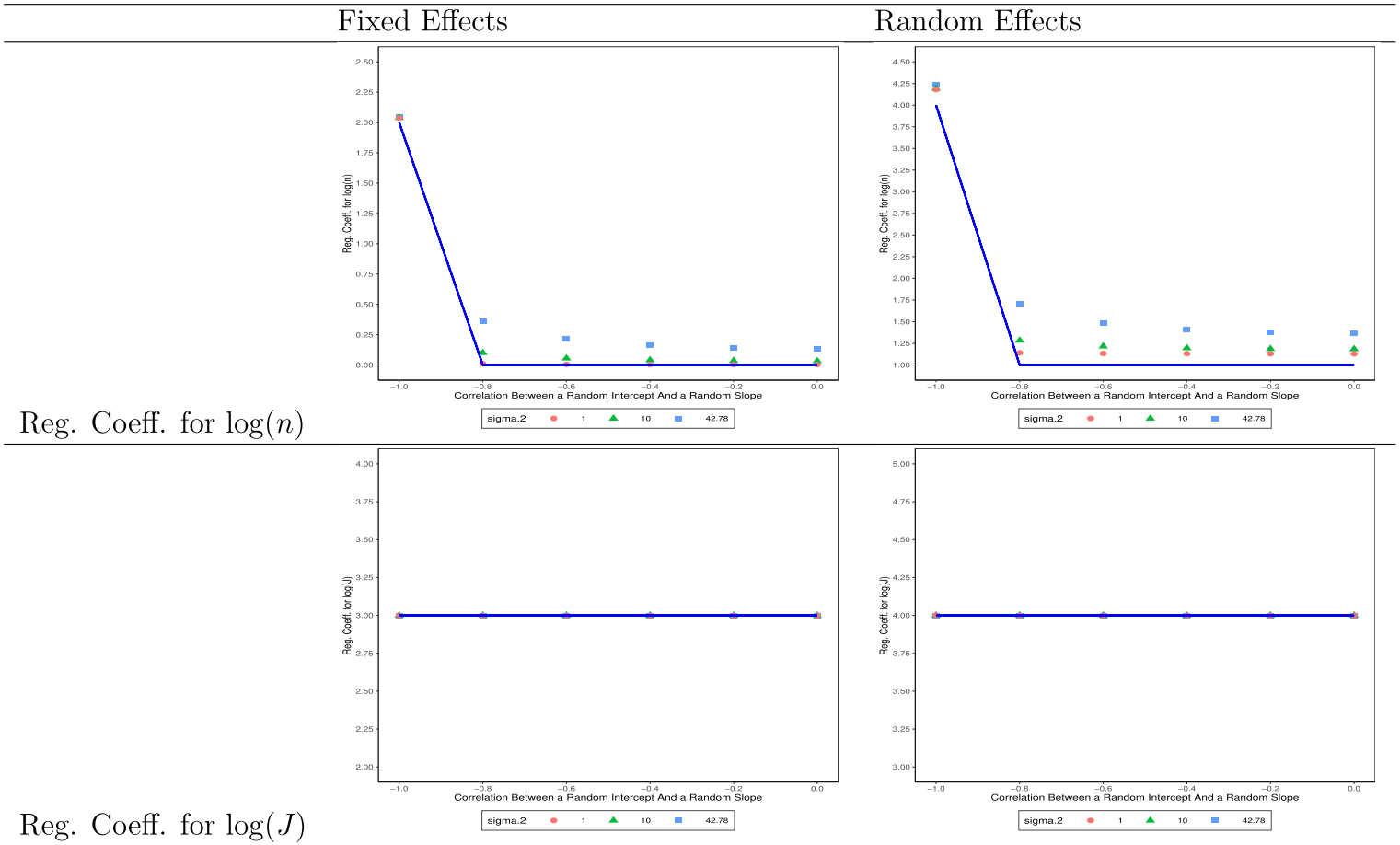


FIGURE 1 Numerical demonstration: Results for Model A (Random-Intercept-Slope Model). Circles (for $\sigma^2 = 1$), triangles (for $\sigma^2 = 10$), and squares (for $\sigma^2 = 42.78$) indicate empirical values; lines indicate expected values.

20448317, 2024, 2, Downloaded from https://bpspsychub.onlinelibrary.wiley.com/doi/10.1111/bmsp.12327 by University Of North Florida, Wiley Online Library on [22/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

the numerical demonstration. For the correlation between b_0 and b_1, a magnitude of -0.5 was selected to examine the behaviour of BIC_{E1}, and a magnitude of -1.0 was selected to examine the behaviour of BIC_{E2}. In Section 5, we showed that the residual variance was an important factor to study the behaviour of BIC_{E1} and BIC_{E2}. Thus, two levels of residual variance, 42.78 and 1, were selected to show the relative behaviour of BIC_{E1} and BIC_{E2} compared to BIC_N and BIC_J. For the data-generating model, different multi-level designs, varying the balanced cluster size n and the number of clusters J, were selected as used in the numerical demonstration: $n = 10, 20$, and 40 and $J = 50, 100, 200$, and 400. To generate the between-cluster and within-cluster covariates, a variable was generated for all observations with a normal distribution with mean 0 and standard deviation 2.07. The between-cluster covariate (x_j) was calculated as the cluster means of this variable, and the within-cluster covariate (x_{ij}) was calculated as the difference between this variable and the cluster means. As the second data-generating model, the random-intercept model with the within-cluster covariate (x_{ij}) was selected to investigate the behaviour of BIC_{E1} in selecting a simpler model. The same true parameters for the simpler model were used as in the random-intercept-slope model (except for β_2, τ_1^2, and τ_{01}).

For each of the two data-generating models, there are four possible fixed-effect structures:

- F1[null]: $\beta_1 = 0, \beta_2 = 0$
- F2[x_{ij}]: $\beta_1 \neq 0, \beta_2 = 0$
- F3[x_j]: $\beta_1 = 0, \beta_2 \neq 0$
- F4[x_{ij}, x_j]: $\beta_1 \neq 0, \beta_2 \neq 0$

In addition, there are two possible random-effect structures:

- V1[random intercept]: $\sigma^2 \neq 0, \tau_0^2 \neq 0, \tau_1^2 = 0$
- V2[random intercept&slope]: $\sigma^2 \neq 0, \tau_0^2 \neq 0, \tau_1^2 \neq 0, \tau_{01} \neq 0$

Note that β_0 exists in all eight models and is considered to be a between-cluster fixed effect known to be non-zero. In total, there are eight models (= 4 fixed-effect structures × 2 random-effect structures). For the random-intercept-slope model, F4 + V2 is the data-generating model and the other seven models are competing models. For the random-intercept model, F2 + V1 is the data-generating model and the other seven models are competing models.

For each condition, 1000 data sets were simulated, and the eight models were fit to the same generated data sets of a data-generating model. As an outcome measure, the percentage of correct model selection (the percentage of replications where the correct model had the lowest BIC) among the eight models was reported to investigate the behaviour of BIC_E (BIC_{E1} or BIC_{E2}), BIC_N, and BIC_J. In addition, for the same simulation results, the percentage of correct model selection among the seven models without a data-generating model was calculated to investigate the behaviour of BIC_E, BIC_N, and BIC_J when the data-generating model was not considered. The 'lmer' function of the `lme4` library (Bates et al., 2015) was used to implement a maximum likelihood estimation method. Table 2 presents a summary of the eight models and the calculations of BIC_{E1} or BIC_{E2}, BIC_N, and BIC_J.

Given the difference in their penalty terms, we would expect BIC_J to have the best performance when the true model was relatively complex among the options and BIC_N to have the best performance when the true model was relatively simple. However, it should be noted that the best selection criterion must achieve a better averaging performance across different levels of complexity of the true model.

20448317, 2024, 2, Downloaded from https://bpspsychub.onlinelibrary.wiley.com/doi/10.1111/bmsp.12327 by University Of North Florida, Wiley Online Library on [22/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

TABLE 2 Simulation study: Summary of calculations of BIC_E, BIC_N, and BIC_J under a random-intercept-slope model as a data-generating model (top) and under a random-intercept model as a data-generating model (bottom).

			BIC			
Model	**Fixed**	**Random**	BIC_E **for cor** $= -0.5$	BIC_E **for cor** $= -1$	BIC_N	BIC_J
Competing	F1[null]	V1[intercept]	$BIC_{E1} = D + 1\log(n) + 3\log(J)$	$BIC_{E1} = D + 1\log(n) + 3\log(J)$	$D + 3\log(N)$	$D + 3\log(J)$
Competing	F1[null]	V2[intercept & slope]	$BIC_{E1} = D + 1\log(n) + 5\log(J)$	$BIC_{E1} = D + 1\log(n) + 5\log(J)$[a]	$D + 5\log(N)$	$D + 5\log(J)$
Competing	F2[x_{ij}]	V1[intercept]	$BIC_{E1} = D + 2\log(n) + 4\log(J)$	$BIC_{E1} = D + 2\log(n) + 4\log(J)$	$D + 4\log(N)$	$D + 4\log(J)$
Competing	F2[x_{ij}]	V2[intercept & slope]	$BIC_{E1} = D + 1\log(n) + 6\log(J)$	$BIC_{E1} = D + 1\log(n) + 6\log(J)$[a]	$D + 6\log(N)$	$D + 6\log(J)$
Competing	F3[x_j]	V1[intercept]	$BIC_{E1} = D + 1\log(n) + 4\log(J)$	$BIC_{E1} = D + 1\log(n) + 4\log(J)$	$D + 4\log(N)$	$D + 4\log(J)$
Competing	F3[x_j]	V2[intercept & slope]	$BIC_{E1} = D + 1\log(n) + 6\log(J)$	$BIC_{E2} = D + 6\log(n) + 6\log(J)$	$D + 6\log(N)$	$D + 6\log(J)$
Competing	F4[x_{ij}, x_j]	V1[intercept]	$BIC_{E1} = D + 2\log(n) + 5\log(J)$	$BIC_{E1} = D + 2\log(n) + 5\log(J)$	$D + 5\log(N)$	$D + 5\log(J)$
Data-Generating	F4[x_{ij}, x_j]	V2[intercept & slope]	$BIC_{E1} = D + 1\log(n) + 7\log(J)$	$BIC_{E2} = D + 6\log(n) + 7\log(J)$	$D + 7\log(N)$	$D + 7\log(J)$

			BIC		
Model	**Fixed**	**Random**	BIC_E	BIC_N	BIC_J
Competing	F1[null]	V1[intercept]	$BIC_{E1} = D + 1\log(n) + 3\log(J)$	$D + 3\log(N)$	$D + 3\log(J)$
Competing	F1[null]	V2[intercept & slope]	$BIC_{E1} = D + 1\log(n) + 5\log(J)$	$D + 5\log(N)$	$D + 5\log(J)$
Data-Generating	F2[x_{ij}]	V1[intercept]	$BIC_{E1} = D + 2\log(n) + 4\log(J)$	$D + 4\log(N)$	$D + 4\log(J)$
Competing	F2[x_{ij}]	V2[intercept & slope]	$BIC_{E2} = D + 5\log(n) + 6\log(J)$[b]	$D + 6\log(N)$	$D + 6\log(J)$
Competing	F3[x_j]	V1[intercept]	$BIC_{E1} = D + 1\log(n) + 4\log(J)$	$D + 4\log(N)$	$D + 4\log(J)$
Competing	F3[x_j]	V2[intercept & slope]	$BIC_{E1} = D + 1\log(n) + 6\log(J)$	$D + 6\log(N)$	$D + 6\log(J)$
Competing	F4[x_{ij}, x_j]	V1[intercept]	$BIC_{E1} = D + 2\log(n) + 5\log(J)$	$D + 5\log(N)$	$D + 5\log(J)$
Competing	F4[x_{ij}, x_j]	V2[intercept & slope]	$BIC_{E2} = D + 5\log(n) + 7\log(J)$[b]	$D + 7\log(N)$	$D + 7\log(J)$

Note: D indicates deviance ($-2\log f(\mathbf{y}|\hat{\boldsymbol{\vartheta}})$).

[a]BIC_{E1} used instead of BIC_{E2} because the random intercept and the random slope are no longer perfectly correlated in order to compensate for the misspecification due to the missing fixed effect.

[b]BIC_{E2} used instead of BIC_{E1} because the variance of the random slope is close to 0 under a random-intercept model as a data-generating model.

20448317, 2024, 2, Downloaded from https://bpspsychub.onlinelibrary.wiley.com/doi/10.1111/bmsp.12327 by University Of North Florida, Wiley Online Library on [22/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

TABLE 3 Simulation study: Percentage of correct model selection by BIC_E, BIC_N, and BIC_J under a random-intercept-slope model as a data-generating model.

	Conditions				Results		
BIC_E	Correlation	Residual variance	n	J	BIC_E	BIC_N	BIC_J
BIC_{E1}	− 0.5	1	10	50	82.2	67	82.2
BIC_{E1}	− 0.5	1	10	100	97.7	93.5	97.7
BIC_{E1}	− 0.5	1	10	200	100	100	100
BIC_{E1}	− 0.5	1	10	400	100	100	100
BIC_{E1}	− 0.5	1	20	50	82.4	61.8	82.4
BIC_{E1}	− 0.5	1	20	100	97.7	89.4	97.7
BIC_{E1}	− 0.5	1	20	200	100	100	100
BIC_{E1}	− 0.5	1	20	400	100	100	100
BIC_{E1}	− 0.5	1	40	50	82.2	54.6	82.2
BIC_{E1}	− 0.5	1	40	100	97.7	89.7	97.7
BIC_{E1}	− 0.5	1	40	200	100	99.9	100
BIC_{E1}	− 0.5	1	40	400	100	100	100
BIC_{E1}	− 0.5	42.78	10	50	80.6	63	80.6
BIC_{E1}	− 0.5	42.78	10	100	96.6	90.8	96.6
BIC_{E1}	− 0.5	42.78	10	200	100	99.8	100
BIC_{E1}	− 0.5	42.78	10	400	100	100	100
BIC_{E1}	− 0.5	42.78	20	50	81.5	59.4	81.5
BIC_{E1}	− 0.5	42.78	20	100	96.8	90.7	96.8
BIC_{E1}	− 0.5	42.78	20	200	100	99.8	100
BIC_{E1}	− 0.5	42.78	20	400	100	100	100
BIC_{E1}	− 0.5	42.78	40	50	82	52.5	82
BIC_{E1}	− 0.5	42.78	40	100	98	88.5	98
BIC_{E1}	− 0.5	42.78	40	200	100	99.9	100
BIC_{E1}	− 0.5	42.78	40	400	100	100	100
BIC_{E2}	− 1	1	10	50	83.2	65.4	83.2
BIC_{E2}	− 1	1	10	100	97.4	93.5	97.4
BIC_{E2}	− 1	1	10	200	100	99.9	100
BIC_{E2}	− 1	1	10	400	100	100	100
BIC_{E2}	− 1	1	20	50	81.5	60.3	81.5
BIC_{E2}	− 1	1	20	100	97.2	90.8	97.2
BIC_{E2}	− 1	1	20	200	100	100	100
BIC_{E2}	− 1	1	20	400	100	100	100
BIC_{E2}	− 1	1	40	50	84.1	56.4	84.1
BIC_{E2}	− 1	1	40	100	97	89.1	97
BIC_{E2}	− 1	1	40	200	100	99.7	100
BIC_{E2}	− 1	1	40	400	100	100	100
BIC_{E2}	− 1	42.78	10	50	82.8	65.9	82.8
BIC_{E2}	− 1	42.78	10	100	97	90.1	97
BIC_{E2}	− 1	42.78	10	200	100	99.9	100

20448317, 2024, 2, Downloaded from https://bpspsychub.onlinelibrary.wiley.com/doi/10.1111/bmsp.12327 by University Of North Florida, Wiley Online Library on [22/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

TABLE 3 (Continued)

	Conditions				Results		
BIC_E	Correlation	Residual variance	n	J	BIC_E	BIC_N	BIC_J
BIC_{E2}	−1	42.78	10	400	100	100	100
BIC_{E2}	−1	42.78	20	50	80.3	56.5	80.3
BIC_{E2}	−1	42.78	20	100	97.6	91.4	97.6
BIC_{E2}	−1	42.78	20	200	100	100	100
BIC_{E2}	−1	42.78	20	400	100	100	100
BIC_{E2}	−1	42.78	40	50	81.8	53.8	81.8
BIC_{E2}	−1	42.78	40	100	97.4	88.2	97.4
BIC_{E2}	−1	42.78	40	200	100	99	100
BIC_{E2}	−1	42.78	40	400	100	100	100

6.2 | Simulation results

Tables 3 and 4 show the percentages of correct model selection by BIC_E (BIC_{E1} or BIC_{E2}), BIC_N, and BIC_J for the random-intercept-slope model (more complex model) and the random-intercept model (simpler model) as data-generating models. A comparison of BIC_E to BIC_N and BIC_J revealed two patterns. First, BIC_E outperformed BIC_J under the simpler data-generating model and performed equally well under the more complex data-generating model, indicating a better overall performance of BIC_E. Second, under the more complex data-generating model, BIC_E had comparable or improved performance compared to BIC_N, while under the simpler data-generating model, BIC_N slightly outperformed BIC_E. Specifically, under the more complex data-generating model, the differences in performance between BIC_E and BIC_N ranged from 15.2% to 29.5% for $J = 50$ and from 3.9% to 9.5% for $J = 100$. Under the simpler data-generating model, the differences in performance were smaller, ranging from 3.4% to 5.4% for $J = 50$ and from 2.1% to 3.1% for $J = 100$. This indicates that overall BIC_E performs better than BIC_N for $J = 50$ and 100. For the two larger numbers of clusters, both BIC_E and BIC_N have near perfect performance. In summary, considering the results in selecting both simpler and more complex models, BIC_E is the best overall selection criterion as it performs at least as well as BIC_N and BIC_J across various multi-level conditions, making it a useful compromise between the two.

When the more complex model was not included as the data-generating model in model selection, a model with similar deviance and penalty (F3[x_j]+V2[intercept&slope]) to those of the data-generating model was selected by all three BICs (BIC_E, BIC_N, and BIC_J) with 100% selection percentage. Similarly, when the simpler model was not considered as the data-generating model in model selection, a model with deviance and penalty (F4[x_{ij}, x_j]+V1[intercept]) similar to those of the data-generating model was selected in BIC_E, BIC_N, and BIC_J, with the highest selection percentage in BIC_E (ranging from 99.8% to 100% across the conditions) followed by BIC_N (ranging from 98.4% to 100% across the conditions) and BIC_J (ranging from 94.4% to 98.1% across the conditions).

7 | EMPIRICAL DATA ILLUSTRATION

In this section, we illustrate the use of the proposed BIC_E using an empirical data set. The empirical data set, `popular.dat`, is from Chapter 2 of Hox (2010) and can be freely downloaded from

20448317, 2024, 2, Downloaded from https://bpspsychub.onlinelibrary.wiley.com/doi/10.1111/bmsp.12327 by University Of North Florida, Wiley Online Library on [22/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

TABLE 4 Simulation study: Percentage of correct model selection by BIC_{E1}, BIC_N, and BIC_J under a random-intercept model as a data-generating model.

Conditions			Results		
Residual variance	n	J	BIC_{E1}	BIC_N	BIC_J
1	10	50	95.1	98.5	94.3
1	10	100	96.0	98.8	95.3
1	10	200	97.6	99.3	97.6
1	10	400	98.2	99.5	98.1
1	20	50	95.1	99.0	93.8
1	20	100	97.3	99.4	97.1
1	20	200	97.8	99.5	97.7
1	20	400	98.8	99.9	98.6
1	40	50	94.1	99.3	93.3
1	40	100	96.6	99.7	96.1
1	40	200	97.7	99.7	97.5
1	40	400	98.2	99.6	98.2
42.78	10	50	95.5	98.9	94.8
42.78	10	100	96.2	98.7	95.7
42.78	10	200	97.8	99.3	97.2
42.78	10	400	98.6	99.4	98.5
42.78	20	50	94.7	98.6	94.1
42.78	20	100	97.1	99.9	96.8
42.78	20	200	97.6	99.8	97.1
42.78	20	400	98.9	100.0	98.7
42.78	40	50	94.3	99.7	93.3
42.78	40	100	97.0	99.6	96.6
42.78	40	200	98.2	99.7	97.8
42.78	40	400	98.4	99.8	98.2

http://joophox.net/mlbook2/DataExchange.zip. The `popular` data set includes 2,000 students from 100 classes ($J = 100$). The average class size is 20 ($\overline{n} = 20$, ranging from 16 to 26, standard deviation= 2.05). Each student belongs to one class, indicating that students (level 1) are nested within classes (level 2). The dependent variable (`popular`) is a self-rated popularity scale ranging from 0 to 10. Within-cluster covariates include a binary-coded gender variable and continuous self-rated extraversion scores. For illustrative purposes, the contrast-coded gender variable ($\texttt{gender}_{ij}$; −1=boy, 1=girl) was included in the following model. The between-cluster covariate chosen was continuous teacher experience in years ($\texttt{texp}_j$). ICC based on an unconditional random-intercept model is 0.365 ($= 0.702/(0.702 + 1.222)$), which means that 36.5% of the variance of popularity scores is at the class level. Thus, the unconditional random-intercept model was considered as the baseline model (Model 1 in Table 5).

We consider a two-level random-intercept-slope model (equation 2.12 in Hox, 2010) based on the following equation:

$$\texttt{popular}_{ij} = \beta_0 + \beta_1 \texttt{gender}_{ij} + \beta_2 \texttt{texp}_j + \beta_3 \texttt{gender}_{ij} \times \texttt{texp}_j + b_{0j} + b_{1j} \texttt{gender}_{ij} + \epsilon_{ij}, \quad (14)$$

20448317, 2024, 2, Downloaded from https://bpspsychub.onlinelibrary.wiley.com/doi/10.1111/bmsp.12327 by University Of North Florida, Wiley Online Library on [22/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

TABLE 5 Empirical Study: BIC_{E1} with comparisons to BIC_N and BIC_J.

Model	Fixed	Random	Par. for K_1	Par. for K_2	Deviance	BIC_{E1}	BIC_N	BIC_J
Model 1	F1[null]	V1[random intercept]	σ^2[1]	β, τ_0^2[2]	6327.5	6344.2(12)	6350.2(12)	6341.2(12)
Model 2	F1[null]	V2[random intercept & slope]	σ^2[1]	$\beta, \tau_0^2, \tau_1^2, \tau_{01}$[4]	5750.5	5776.4(10)	5788.4(10)	5773.4(10)
Model 3	F2[gender]	V1[random intercept]	β_1, σ^2[2]	β, τ_0^2[2]	5556.3	5580.6(5)	5586.6(5)	5574.6(5)
Model 4	F2[gender]	V2[random intercept & slope]	σ^2[1]	$\beta, \beta_1, \tau_0^2, \tau_1^2, \tau_{01}$[5]	5551.5	5582.0(6)	5597.0(7)	5579.0(7)
Model 5	F3[gender,gender×texp]	V1[random intercept]	$\beta_1, \beta_3, \sigma^2$[3]	β, τ_0^2[2]	5552.1	5584.0(7)	5590.0(6)	5575.0(6)
Model 6	F3[gender,gender×texp]	V2[random intercept & slope]	σ^2[1]	$\beta, \beta_1, \beta_3, \tau_0^2, \tau_1^2, \tau_{01}$[6]	5549.5	5584.8(8)	5602.8(8)	5581.8(8)
Model 7	F4[texp]	V1[random intercept]	σ^2[1]	β, β_2, τ_0^2[3]	6303.0	6324.4(11)	6333.4(11)	6321.4(11)
Model 8	F4[texp]	V2[random intercept & slope]	σ^2[1]	$\beta, \beta_2, \tau_0^2, \tau_1^2, \tau_{01}$[5]	5730.0	5760.6(9)	5775.6(9)	5757.6(9)
Model 9	F5[gender,texp]	V1[random intercept]	β_1, σ^2[2]	β, β_2, τ_0^2[3]	5528.5	5557.4(1)	5566.4(1)	5551.4(2)
Model 10	F5[gender,texp]	V2[random intercept & slope]	σ^2[1]	$\beta, \beta_1, \beta_2, \tau_0^2, \tau_1^2, \tau_{01}$[6]	5524.9	5560.0(3)	5578.0(3)	5557.0(3)
Model 11	F6[gender,texp,gender×texp]	V1[random intercept]	$\beta_1, \beta_3, \sigma^2$[3]	β, β_2, τ_0^2[3]	5523.4	5560.0(2)	5569.0(2)	5551.0(1)
Model 12	F6[gender,texp,gender×texp]	V2[random intercept & slope]	σ^2[1]	$\beta, \beta_1, \beta_2, \beta_3, \tau_0^2, \tau_1^2, \tau_{01}$[7]	5520.6	5560.4(4)	5581.4(4)	5557.4(4)

Note: Numbers in parentheses indicate rank order of BIC_{E1}, BIC_N, and BIC_J from smallest to largest; numbers in square brackets for columns of par. (parameters) for K_1 and par. for K_2 indicate K_1 and K_2 respectively.

20448317, 2024, 2, Downloaded from https://bpspsychub.onlinelibrary.wiley.com/doi/10.1111/bmsp.12327 by University Of North Florida, Wiley Online Library on [22/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

where β_0, β_1, β_2, and β_3 are the fixed effects of intercept, within-cluster covariate $\texttt{gender}_{ij}$, between-cluster covariate $\texttt{texp}_j$, and their cross-level interaction respectively, and b_{0j} and b_{1j} are the random intercept and the random slope of the within-cluster covariate $\texttt{gender}_{ij}$, and ϵ_{ij} is the random residual. We assume $\epsilon_{ij} \sim \mathrm{N}(0, \sigma^2)$ and $(b_{0j}, b_{1j})' \sim \mathrm{MVN}(\mathbf{0}, \Psi)$, where the variances of b_{0j} and b_{1j} are τ_0^2 and τ_1^2, and their covariance is τ_{01}.

For the two-level random-intercept-slope model, there are six possible fixed-effect structures to be compared:

- F1[null]: $\beta_1 = 0, \beta_2 = 0, \beta_3 = 0$
- F2[gender]: $\beta_1 \neq 0, \beta_2 = 0, \beta_3 = 0$
- F3[gender,gender×texp]: $\beta_1 \neq 0, \beta_2 = 0, \beta_3 \neq 0$
- F4[texp]: $\beta_1 = 0, \beta_2 \neq 0, \beta_3 = 0$
- F5[gender,texp]: $\beta_1 \neq 0, \beta_2 \neq 0, \beta_3 = 0$
- F6[gender,texp,gender×texp]: $\beta_1 \neq 0, \beta_2 \neq 0, \beta_3 \neq 0$

In addition, there are two possible random-effect structures to be compared:

- V1[random intercept]: $\sigma^2 \neq 0, \tau_0^2 \neq 0, \tau_1^2 = \tau_{01} = 0$
- V2[random intercept & slope]: $\sigma^2 \neq 0, \tau_0^2 \neq 0, \tau_1^2 \neq 0, \tau_{01} \neq 0$

The objective of the current analysis is to simultaneously identify the important covariates that correspond to the level-specific fixed and random effects among the 12 candidate models (6 fixed-effect structures × 2 random-effect structures, summarized in Table 5) based on BIC_E. Note that the intercept exists in all 12 models but is known to be non-zero. In Table 5, parameters counting toward K_1 and K_2 are listed for illustrative purposes. The following patterns should be noted. First, the fixed intercept parameter (β_0) counts toward K_2 because it is associated with a random-intercept parameter. Second, the fixed effect of the within-cluster covariate gender_{ij} (β_1) and its interaction effect with texp_j (β_3) count toward K_1 in the random-intercept model because their two columns in the design matrix $\mathbf{X}_j$ are not in the column space of the design matrix $\mathbf{Z}_j$, which has a single column of 1's; however, they count toward K_2 when the random slope of gender_{ij} is present because their two columns in $\mathbf{X}_j$ are now multiples of the column of $\mathbf{Z}_j$ for gender_{ij}. Third, the fixed effect of the between-cluster covariate texp_j is always counted for K_2 as its column in $\mathbf{X}_j$ is a vector of identical values.

The BIC is based on the deviance (−2 × log-likelihood). The 'lmer' function of the `lme4` library was used to fit the 12 models and to obtain their deviances. The maximum likliehood estimation method (Goldstein, 1986; Longford, 1987) is considered to be the most appropriate estimation method for information criteria (Verbeke & Molenberghs, 2000). The problem with the maximum likliehood estimation method is that it tends to underestimate variance components. This tendency is not limited to the residual variance and gets worse as the number of fixed effects increases (e.g., Verbeke & Molenberghs, 2000). In such a case, restricted maximum likelihood (REML; Patterson & Thompson, 1971) is recommended. In the current applications, for all models we considered, the deviance from maximum liklihood and restricted maximum likelihood estimation methods was the same, and variance estimates from maximum liklihood and restricted maximum likelihood estimation methods differed in the second or third decimal point. Therefore, the deviance from maximum likelihood estimation was used in calculating BICs. There was no empirical evidence of rank deficiency (such as random-effect variances close to 0) in all 12 models or correlations of a random intercept and a random slope near 1 or −1 in Models 2, 4, 6, 8, 10, and 12. Thus, the BIC_{E1} (instead of BIC_{E2}) is compared with BIC_N and BIC_J.

20448317, 2024, 2, Downloaded from https://bpspsychub.onlinelibrary.wiley.com/doi/10.1111/bmsp.12327 by University Of North Florida, Wiley Online Library on [22/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

20448317, 2024, 2, Downloaded from https://bpspsychub.onlinelibrary.wiley.com/doi/10.1111/bmsp.12327 by University Of North Florida, Wiley Online Library on [22/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

As shown in Table 5, the 12 candidate models were ranked differently across BIC_{E1}, BIC_N, and BIC_J for Models 4 and 5 and Models 9–12. For Models 4 and 5, BIC_{E1} increased from Model 4 (F2+V2) to Model 5 (F3+V1), whereas BIC_N and BIC_J decreased from Model 4 to Model 5. For Models 9–12, all three BICs increased from the random-intercept models (V1) to the random intercept and slope models (V2). However, they suggested different fixed models: BIC_{E1} and BIC_N suggested Model 9 (F5+V1), whereas BIC_J suggested Model 11 (F6+V1). This example presents different possible results by different kinds of BIC (BIC_{E1}, BIC_N, and BIC_J).

8 | SUMMARY AND DISCUSSION

BIC is provided in popular statistical software for MLM. However, no consensus exists yet in the literature or in popular software on the calculation of BIC for MLM, posing a key problem in the selection of fixed and random effects. In this paper, we derived the BIC's penalty term for both fixed-effect and random-effect parameters in a two-level nested model in its most general form. The proposed BIC_{E1} and BIC_{E2} can be used to select among models that differ in level-specific fixed and random effects in MLM applications.

In the derived BIC_{E1} and BIC_{E2}, the total penalty term in BIC is decomposed into two terms: (a) a term with the log of average sample size per cluster whose multiplier involves the overlapping number of dimensions between the column spaces of the random- and fixed-effect design matrices and (b) the total number of parameters times the log of the total number of clusters. This result is consistent with Pauler's (1998) results for selecting among models that differ only in their fixed-effect parameters in a random intercept model but is more general in its inclusion of random-effect parameters. This result is also consistent with Delattre et al.'s (2014) results derived under a more specific form of a random-effect model.

In addition, we studied the situation where a variance–covariance matrix of random effects is singular, giving an analytical expression of the coefficient of $\log(n)$ in BIC_{E2} in this situation. We found that in this situation, the coefficient is greater than its desired value with a non-singular random-effect variance–covariance matrix. The derived BIC_{E2} in this study is particularly useful when comparing models that differ not only in the number of fixed effects but also in the number of random effects, in which case some models typically have *near*-singular estimated variance–covariance matrices of random effects.

The aim of this study was to provide the analytical derivation of the BIC's penalty term in the joint selection of level-specific fixed effects and random effects in a two-level nested design. Through a numerical demonstration, BIC_E (BIC_{E1} and BIC_{E2}) was found to behave consistently with our theoretical predictions. Additionally, we showed through a simulation study that BIC_E, BIC_N, and BIC_J behaved differently depending on which data-generating model was used, except when the cluster size and the number of clusters was very large. BIC_N was not consistently satisfactory in selecting a simpler or more complex model as data-generating models. Moreover, BIC_E outperformed BIC_J when the cluster size and the number of clusters was relatively small and the data-generating model was simpler, but performed equally well otherwise. Considering the overall performances of these BICs under both simpler and more complex data-generating models, BIC_E is recommended over BIC_N and BIC_J, especially when the number of clusters is relatively small. As with all simulation studies, our findings are limited to the conditions we considered, and additional simulation studies may be needed to generalize these findings to other conditions.

Furthermore, the use of BIC_{E1} was illustrated using a textbook example for selecting among candidate models that have different level-specific fixed and random effects (with a full-rank variance–covariance matrix). As illustrated, BIC_{E1} can be easily calculated using deviance from software and by counting K_1, K_2, and sample sizes.

This study presented BIC_{E1} and BIC_{E2} for the two-level nested data. It is left for a future study to further explore the forms of BIC in higher-level nested designs and for other complex multi-level designs (e.g., a cross-classified design, a longitudinal design with correlated errors). The form of BIC is tied to the asymptotic behaviour of the log-determinant of the information matrix, which is a more complicated form of these designs.

As many studies on BIC (e.g., Delattre et al., 2014; Jones, 2011; Pauler, 1998), this study considers BIC as an asymptotic approximation of the marginal probability of the observed data, so that the difference in BIC between two models properly approximates the logarithm of Bayes factor. In the simulation study, we demonstrated empirically that the BIC_{E1} and BIC_{E2} selected the data-generating model as a true model asymptotically. However, we did not study their analytical properties in model selection, such as consistency, which can be a topic for future studies.

AUTHOR CONTRIBUTIONS

Sun-Joo Cho: conceptualization; data curation; formal analysis; investigation; methodology; project administration; software; validation; visualization; writing – original draft. **Hao Wu:** conceptualization; data curation; formal analysis; investigation; methodology; software; validation; writing – original draft. **Matthew Naveiras:** writing – review and editing.

CONFLICT OF INTEREST STATEMENT

The authors have no conflicts of interest to declare that are relevant to the content of this article.

DATA AVAILABILITY STATEMENT

The empirical data set, popular.dat, which is utilized in this article, is sourced from Chapter 2 of Hox (2010). It can be freely downloaded from http://joophox.net/mlbook2/DataExchange.zip.

ORCID

Sun-Joo Cho https://orcid.org/0000-0002-2600-8305
Hao Wu https://orcid.org/0000-0001-6471-1774
Matthew Naveiras https://orcid.org/0000-0002-6783-0274

REFERENCES

Baayen, R. H., Davidson, D. J., & Bates, D. M. (2008). Mixed-effects modeling with crossed random effects for subjects and items. *Journal of Memory and Language*, *59*, 390–412. https://doi.org/10.1016/j.jml.2007.12.005

Barr, D. J., Levy, R., Scheepers, C., & Tily, H. J. (2013). Random effects structure for confirmatory hypothesis testing: Keep it maximal. *Journal of Memory and Language*, *68*, 255–278. https://doi.org/10.1016/j.jml.2012.11.001

Bates, D., Mächler, M., Bolker, B., & Walker, S. (2015). Fitting linear mixed-effects models using lme4. *Journal of Statistical Software*, *67*, 1–48. https://doi.org/10.1016/S0378-3758(02)00336-1

Berger, J., Ghosh, J. K., & Mukhopadhyay, N. (2003). Approximations and consistency of Bayes factors as model dimension grows. *Journal of Statistical Planning and Inference*, *112*, 241–258. https://doi.org/10.1016/S0378-3758(02)00336-1

Burnham, K. P., & Anderson, D. R. (2002). *Model selection and multimodel inference: A practical information-theoretic approach* (2nd ed.). Springer. https://doi.org/10.1007/b97636

Claeskens, G., & Hjort, N. (2008). *Model selection and model averaging*. Cambridge University Press. https://doi.org/10.1017/CBO9780511790485

de Bruijn, N. G. (1970). *Asymptotic methods in analysis*. North-Holland.

de Leeuw, J., & Meijer, E. (2007). Introduction to multilevel analysis. In J. de Leeuw & E. Meijer (Eds.), *Handbook of advanced multilevel analysis* (pp. 1–75). Springer. https://doi.org/10.4324/9780203848852

Delattre, M., Lavielle, M., & Poursat, M. A. (2014). A note on BIC in mixed-effects models. *Electronic Journal of Statistics*, *8*, 456–475. https://doi.org/10.1214/14-EJS890

Goldstein, H. (1986). Multilevel mixed linear model analysis using iterative generalized least squares. *Biometrika*, *73*, 43–56. https://doi.org/10.1093/biomet/73.1.43

Goldstein, H. (2003). *Multilevel statistical models* (3rd ed.). Oxford University Press [Distributor]. https://doi.org/10.1002/9780470973394

20448317, 2024, 2, Downloaded from https://bpspsychub.onlinelibrary.wiley.com/doi/10.1111/bmsp.12327 by University Of North Florida, Wiley Online Library on [22/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Hamaker, E. L., van Hattum, P., Kuiper, R. M., & Hoijtink, H. (2011). Model selection based on information criteria in multilevel modeling. In J. J. Hox & J. K. Roberts (Eds.), *Handbook for advanced multilevel analysis* (pp. 231–255). Routledge/Taylor & Francis Group. https://doi.org/10.4324/9780203848852

Hox, J. J. (2010). *Multilevel analysis: Techniques and applications* (2nd ed.). IEA. https://doi.org/10.4324/9780203852279

Hox, J. J., Moerbeek, M., & van de Schoot, R. (2018). *Multilevel analysis: Techniques and applications* (3rd ed.). Routledge. https://doi.org/10.4324/9781315650982

IBM Corp. (2020). *IBM SPSS Statistics for Windows, Version 27.0*. IBM Corp.

Jones, R. H. (2011). Bayesian information criterion for longitudinal and clustered data. *Statistics in Medicine*, *30*, 3050–3056. https://doi.org/10.1080/01621459.1995.10476572

Kass, R. E., & Raftery, A. E. (1995). Bayes factors. *Journal of the American Statistical Association*, *90*, 773–795. https://doi.org/10.1080/01621459.1995.10476572

Kass, R. E., & Vaidyanathan, S. K. (1992). Approximate Bayes factors and orthogonal parameters, with application to testing equality of two Binomial proportions. *Journal of the Royal Statistical Society. Series B, Methodological*, *54*(1), 129–144. https://doi.org/10.1111/j.2517-6161.1992.tb01868.x

Kreft, I. G., & de Leeuw, J. (2002). *Introducing multilevel modeling*. SAGE Publications, Ltd. https://doi.org/10.4135/9781849209366

Laird, N. M., & Ware, J. H. (1982). Random-effects models for longitudinal data. *Biometrics*, *38*, 963–974. https://doi.org/10.2307/2529876

Longford, N. T. (1987). A fast scoring algorithm for maximum likelihood estimation in unbalanced mixed models with nested random effects. *Biometrika*, *74*, 817–827. https://doi.org/10.1093/biomet/74.4.817

Lorah, J., & Womack, A. (2019). Value of sample size for computation of the Bayesian information criterion (BIC) in multilevel modeling. *Behavior Research Methods*, *51*, 440–450. https://doi.org/10.3758/s13428-018-1188-3

McCoach, D. B., & Black, A. C. (2008). Evaluation of model fit and adequacy. In A. A. O'Connell & D. B. McCoach (Eds.), *Multilevel modeling of educational data* (pp. 245–272). Information Age Publishing, Inc.

Muthén, L. K., & Muthén, B. O. (1998–2015). *Mplus user's guide* (7th ed.). Muthén & Muthén. Retrieved from https://www.statmodel.com/download/usersguide/Mplus%20user%20guide%20Ver_7_r3_web.pdf

Patterson, H. D., & Thompson, R. (1971). Recovery of inter-block information when block sizes are unequal. *Biometrika*, *58*, 545–554. https://doi.org/10.2307/2334389

Pauler, D. K. (1998). The Schwarz criterion and related methods for normal linear models. *Biometrika*, *85*, 13–27. https://doi.org/10.1093/biomet/85.1.13

Raftery, A. E. (1995). Bayesian model selection in social research. *Sociological Methodology*, *25*, 111–163. https://doi.org/10.2307/271063

SAS Institute Inc. (2015). *SAS/STAT® 14.1 user's guide*. SAS Institute Inc.

Schwarz, G. (1978). Estimating the dimension of a model. *Annals of Statistics*, *6*, 461–464. https://doi.org/10.1214/aos/1176344136

Shen, N., & González, B. (2021). Bayesian information criterion for linear mixed-effects models. Retrieved June 25, 2022, from https://arxiv.org/abs/2104.14725

Snijders, T. A. B., & Bosker, R. J. (2012). *Multilevel analysis: An introduction to basic and advanced multilevel modeling* (2nd ed.). Sage. https://doi.org/10.1080/10705511.2013.797841

Stone, M. (1979). Comments on model selection criteria of Akaike and Schwarz. *Journal of the Royal Statistical Society (Series B)*, *41*, 276–278. https://doi.org/10.1111/j.2517-6161.1979.tb01084.x

Tierney, L., & Kadane, J. B. (1986). Accurate approximations for posterior moments and marginal densities. *Journal of the American Statistical Association*, *81*, 82–86. https://doi.org/10.1080/01621459.1986.10478240

Verbeke, G., & Molenberghs, G. (2000). *Linear mixed models for longitudinal data*. Springer. https://doi.org/10.1007/978-1-4419-0300-6

Whittaker, T. A., & Furlow, C. F. (2009). The comparison of model selection criteria when selecting among competing hierarchical linear models. *Journal of Modern Applied Statistical Methods*, *8*, 173–193. https://doi.org/10.22237/jmasm/1241136840

Wolfinger, R. (1993). Laplace's approximation for nonlinear mixed models. *Biometrika*, *80*, 791–795. https://doi.org/10.1093/biomet/80.4.791

How to cite this article: Cho, S.-J., Wu, H., & Naveiras, M. (2024). The effective sample size in Bayesian information criterion for level-specific fixed and random-effect selection in a two-level nested model. *British Journal of Mathematical and Statistical Psychology*, *77*, 289–315. https://doi.org/10.1111/bmsp.12327

20448317, 2024, 2, Downloaded from https://bpspsychub.onlinelibrary.wiley.com/doi/10.1111/bmsp.12327 by University Of North Florida, Wiley Online Library on [22/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

APPENDIX A

DERIVATION OF BIC WHEN RANDOM-EFFECT COVARIANCE MATRIX HAS FULL RANK

A.1 | Information matrix of fixed effects

In this section we study the asymptotic behaviour of $\log|\mathbf{I}_{\boldsymbol{\beta\beta}}|$. Noting that $\mathbf{I}_{\boldsymbol{\beta\beta}} = \sum_j \mathbf{I}_{\boldsymbol{\beta\beta},j}$, we first derive the form of $\mathbf{I}_{\boldsymbol{\beta\beta},j}$. We define

$$\mathbf{S}_{XX,n_j} = \mathbf{X}_j'\mathbf{X}_j/n_j, \qquad \mathbf{S}_{XZ,n_j} = \mathbf{X}_j'\mathbf{Z}_j/n_j, \qquad \mathbf{S}_{ZZ,n_j} = \mathbf{Z}_j'\mathbf{Z}_j/n_j.$$

Note that while the number of rows of $\mathbf{X}_j$ and $\mathbf{Z}_j$ grows with n_j, the sizes of the matrices defined above remain constant and their elements are assumed to remain bounded.

The block of cluster j's information matrix $\mathbf{I}_j$ corresponding to the fixed effects $\boldsymbol{\beta}$ takes the following form (also see Pauler, 1998):

$$\mathbf{I}_{\boldsymbol{\beta\beta},j} = \mathbf{X}_j'\mathbf{V}_j^{-1}\mathbf{X}_j. \tag{A1}$$

Here we note that (also see Pauler, 1998)

$$\mathbf{V}_j^{-1} = \frac{1}{\sigma^2}\left\{I_{n_j} - \mathbf{Z}_j(\sigma^2\Psi^{-1} + \mathbf{Z}_j'\mathbf{Z}_j)^{-1}\mathbf{Z}_j'\right\} = \frac{1}{\sigma^2}\left\{I_{n_j} - \frac{1}{n_j}\mathbf{Z}_j\left[\frac{\sigma^2}{n_j}\Psi^{-1} + \mathbf{S}_{ZZ,n_j}\right]^{-1}\mathbf{Z}_j'\right\}$$

because

$$\begin{aligned}
&\left[\sigma^2(n_j\Psi)^{-1} + \mathbf{S}_{ZZ,n_j}\right]^{-1} - \mathbf{S}_{ZZ,n_j}^{-1} \\
&= \left[\sigma^2(n_j\Psi)^{-1} + \mathbf{S}_{ZZ,n_j}\right]^{-1}\mathbf{S}_{ZZ,n_j}\mathbf{S}_{ZZ,n_j}^{-1} - \left[\sigma^2(n_j\Psi)^{-1} + \mathbf{S}_{ZZ,n_j}\right]^{-1}\left[\sigma^2(n_j\Psi)^{-1} + \mathbf{S}_{ZZ,n_j}\right]\mathbf{S}_{ZZ,n_j}^{-1} \\
&= -\left[\sigma^2(n_j\Psi)^{-1} + \mathbf{S}_{ZZ,n_j}\right]^{-1}\sigma^2(n_j\Psi)^{-1}\mathbf{S}_{ZZ,n_j}^{-1} \\
&= -\frac{\sigma^2}{n_j}\mathbf{S}_{ZZ,n_j}^{-1}\left[\frac{\sigma^2}{n_j}\mathbf{S}_{ZZ,n_j}^{-1} + \Psi\right]^{-1}\mathbf{S}_{ZZ,n_j}^{-1}.
\end{aligned}$$

As a result, we have

$$\mathbf{V}_j^{-1} = \frac{1}{\sigma^2}\left\{I_{n_j} - \frac{1}{n_j}\mathbf{Z}_j\mathbf{S}_{ZZ,n_j}^{-1}\mathbf{Z}_j'\right\} + \frac{1}{n_j^2}\mathbf{Z}_j\mathbf{S}_{ZZ,n_j}^{-1}\left[\frac{\sigma^2}{n_j}\mathbf{S}_{ZZ,n_j}^{-1} + \Psi\right]^{-1}\mathbf{S}_{ZZ,n_j}^{-1}\mathbf{Z}_j',$$

and Equation (A1) becomes

$$\mathbf{I}_{\boldsymbol{\beta\beta},j} = \frac{n_j}{\sigma^2}\left\{\mathbf{S}_{XX,n_j} - \mathbf{S}_{XZ,n_j}\mathbf{S}_{ZZ,n_j}^{-1}\mathbf{S}_{ZX,n_j}\right\} + \mathbf{S}_{XZ,n_j}\mathbf{S}_{ZZ,n_j}^{-1}\left[\frac{\sigma^2}{n_j}\mathbf{S}_{ZZ,n_j}^{-1} + \Psi\right]^{-1}\mathbf{S}_{ZZ,n_j}^{-1}\mathbf{S}_{ZX,n_j}.$$

20448317, 2024, 2, Downloaded from https://bpspsychub.onlinelibrary.wiley.com/doi/10.1111/bmsp.12327 by University Of North Florida, Wiley Online Library on [22/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

The first term of this expression suggests that we should decompose $\mathbf{X}_j$ into two terms, one in the column space of $\mathbf{Z}_j$ and the other orthogonal to it. We write

$$\mathbf{X}_j = \mathbf{Z}_j\mathbf{W}_j + \mathbf{E}_j$$

for some $q \times p$ matrix $\mathbf{W}_j$ and some $n_j \times p$ matrix $\mathbf{E}_j$ with $\mathbf{E}_j'\mathbf{Z}_j = \mathbf{O}$ (a $p \times q$ zero matrix). This leads to

$$\mathbf{I}_{\boldsymbol{\beta}\boldsymbol{\beta},j} = \frac{n_j}{\sigma^2}\mathbf{S}_{EE,n_j} + \mathbf{W}_j'\left[\frac{\sigma^2}{n_j}\mathbf{S}_{ZZ,n_j}^{-1} + \boldsymbol{\Psi}\right]^{-1}\mathbf{W}_j,$$

where $\mathbf{S}_{EE,n_j} = \mathbf{E}_j'\mathbf{E}_j/n_j$. Summing over clusters, we have

$$\mathbf{I}_{\boldsymbol{\beta}\boldsymbol{\beta}} = \sum_j \mathbf{I}_{\boldsymbol{\beta}\boldsymbol{\beta},j} = \frac{1}{\sigma^2}\sum_j n_j\mathbf{S}_{EE,n_j} + \sum_j \mathbf{W}_j'\left[\frac{\sigma^2}{n_j}\mathbf{S}_{ZZ,n_j}^{-1} + \boldsymbol{\Psi}\right]^{-1}\mathbf{W}_j. \tag{A2}$$

We now study the order of $\mathbf{I}_{\boldsymbol{\beta}\boldsymbol{\beta}}$. The summand in the second term of Equation (A2) has eigenvalues that are bounded away from zero. For the first term, assume that $\overline{\mathbf{S}}_{EE} = \frac{1}{N}\sum_j n_j\mathbf{S}_{EE,n_j}$ is bounded and has (limiting) rank p_1, and let $p_2 = p - p_1$. We have

$$\log|\mathbf{I}_{\boldsymbol{\beta}\boldsymbol{\beta}}| = p_2\log(J) + p_1\log(N) + O_p(1) = p\log(J) + p_1\log(\overline{n}) + O_p(1), \tag{A3}$$

where $\overline{n}$ is the average cluster size.

A.2 | Information matrix of random-effect parameters

In this section we study the asymptotic behaviour of $\log|\mathbf{I}_{\boldsymbol{\rho}\boldsymbol{\rho}}|$. Recall that $\boldsymbol{\rho} = [\mathrm{vech}'(\boldsymbol{\Psi}), \sigma^2]'$, so the information matrix $\mathbf{I}_{\boldsymbol{\rho}\boldsymbol{\rho},j}$ for cluster j has three non-duplicated blocks. We start with the element corresponding to σ^2. It is given by

$$\mathbf{I}_{\sigma^2\sigma^2,j} = \frac{1}{2}\mathrm{tr}\left\{\frac{\partial\mathbf{V}_j}{\partial\sigma^2}\mathbf{V}_j^{-1}\frac{\partial\mathbf{V}_j}{\partial\sigma^2}\mathbf{V}_j^{-1}\right\} = \frac{1}{2}\mathrm{tr}(\mathbf{V}_j^{-2}).$$

Here we note

$$\mathbf{V}_j^{-2} = \frac{1}{\sigma^4}\left\{I_{n_j} - \frac{1}{n_j}\mathbf{Z}_j\mathbf{S}_{ZZ,n_j}^{-1}\mathbf{Z}_j'\right\} + \frac{1}{n_j^3}\mathbf{Z}_j\mathbf{S}_{ZZ,n_j}^{-1}\left[\frac{\sigma^2}{n_j}\mathbf{S}_{ZZ,n_j}^{-1} + \boldsymbol{\Psi}\right]^{-1}\mathbf{S}_{ZZ,n_j}^{-1}\left[\frac{\sigma^2}{n_j}\mathbf{S}_{ZZ,n_j}^{-1} + \boldsymbol{\Psi}\right]^{-1}\mathbf{S}_{ZZ,n_j}^{-1}\mathbf{Z}_j',$$

20448317, 2024, 2, Downloaded from https://bpspsychub.onlinelibrary.wiley.com/doi/10.1111/bmsp.12327 by University Of North Florida, Wiley Online Library on [22/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

from which we obtain

$$\mathbf{I}_{\sigma^2\sigma^2,j} = \frac{n_j - q}{2\sigma^4} + \frac{1}{2n_j^2}\mathrm{tr}\left\{\left[\frac{\sigma^2}{n_j}I_q + \Psi\mathbf{S}_{ZZ,n_j}\right]^{-2}\right\}.$$

Note that this term is always of order $O(n_j)$, so $\mathbf{I}_{\sigma^2\sigma^2} = \sum_j \mathbf{I}_{\sigma^2\sigma^2,j} = O(N)$.

The diagonal block of $\mathbf{I}_{\boldsymbol{\rho}\boldsymbol{\rho},j}$ corresponding to $\boldsymbol{\psi} = \mathrm{vech}(\Psi)$ is given by

$$\begin{aligned}
\mathbf{I}_{\boldsymbol{\psi}\boldsymbol{\psi},j} &= \frac{1}{2}\left[\frac{\partial \mathrm{vec}\mathbf{V}_j}{\partial \boldsymbol{\psi}'}\right]'\left\{\mathbf{V}_j^{-1}\otimes\mathbf{V}_j^{-1}\right\}\left[\frac{\partial \mathrm{vec}\mathbf{V}_j}{\partial \boldsymbol{\psi}'}\right] \\
&= \frac{1}{2}\mathbf{D}_q'\left\{\mathbf{Z}_j\otimes\mathbf{Z}_j\right\}'\left\{\mathbf{V}_j^{-1}\otimes\mathbf{V}_j^{-1}\right\}\left\{\mathbf{Z}_j\otimes\mathbf{Z}_j\right\}\mathbf{D}_q \\
&= \frac{1}{2}\mathbf{D}_q'\left\{\left[\mathbf{Z}_j'\mathbf{V}_j^{-1}\mathbf{Z}_j\right]\otimes\left[\mathbf{Z}_j'\mathbf{V}_j^{-1}\mathbf{Z}_j\right]\right\}\mathbf{D}_q \\
&= \frac{1}{2}\mathbf{D}_q'\left\{\left[\Psi + \frac{\sigma^2}{n_j}\mathbf{S}_{ZZ,n_j}^{-1}\right]^{-1}\otimes\left[\Psi + \frac{\sigma^2}{n_j}\mathbf{S}_{ZZ,n_j}^{-1}\right]^{-1}\right\}\mathbf{D}_q,
\end{aligned}$$

where $\mathbf{D}_q$ is the $q^2 \times \frac{q(q+1)}{2}$ duplication matrix and vec vectorizes a full matrix by connecting its columns. This matrix has constant order, so $\mathbf{I}_{\boldsymbol{\psi}\boldsymbol{\psi}} = \sum_j \mathbf{I}_{\boldsymbol{\psi}\boldsymbol{\psi},j} = O_p(J)$.

The off-diagonal block is

$$\begin{aligned}
\mathbf{I}_{\boldsymbol{\psi}\sigma^2,j} &= \frac{1}{2}\left[\frac{\partial \mathrm{vec}\mathbf{V}_j}{\partial \boldsymbol{\psi}'}\right]'\left\{\mathbf{V}_j^{-1}\otimes\mathbf{V}_j^{-1}\right\}\left[\frac{\partial \mathrm{vec}\mathbf{V}_j}{\partial \sigma^2}\right] \\
&= \frac{1}{2}\mathbf{D}_q'\left\{\mathbf{Z}_j\otimes\mathbf{Z}_j\right\}'\left\{\mathbf{V}_j^{-1}\otimes\mathbf{V}_j^{-1}\right\}\left[\mathrm{vec}\mathbf{I}_{n_j}\right] \\
&= \frac{1}{2}\mathbf{D}_q'\mathrm{vec}(\mathbf{Z}_j'\mathbf{V}^{-2}\mathbf{Z}_j) \\
&= \frac{1}{2n_j}\mathbf{D}_q'\mathrm{vec}\left\{\left[\frac{\sigma^2}{n_j}I_q + \mathbf{S}_{ZZ,n_j}\Psi\right]^{-2}\mathbf{S}_{ZZ,n_j}\right\}.
\end{aligned}$$

This off-diagonal block has order $o_p(1)$, so $\mathbf{I}_{\boldsymbol{\psi}\sigma^2} = \sum_j \mathbf{I}_{\boldsymbol{\psi}\sigma^2,j} = o_p(J)$.

The information matrix $\mathbf{I}_{\boldsymbol{\rho}\boldsymbol{\rho}}$ of random-effect parameters satisfies

$$\log|\mathbf{I}_{\boldsymbol{\rho}\boldsymbol{\rho}}| = q^*\log(J) + \log(N) + O_p(1) = (q^*+1)\log(J) + \log(\overline{n}) + O_p(1), \tag{A4}$$

where $q^* = q(q+1)/2$ is the number of non-duplicated elements in Ψ.

APPENDIX B

DERIVATION OF BIC WHEN A REDUNDANT RANDOM EFFECT IS PRESENT

B.1 | Notations

When Ψ is singular, some of the random effects are redundant. We only study the simplified situation where some of the random effects have zero variance and the remaining random effects have a

20448317, 2024, 2, Downloaded from https://bpspsychub.onlinelibrary.wiley.com/doi/10.1111/bmsp.12327 by University Of North Florida, Wiley Online Library on [22/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

20448317, 2024, 2, Downloaded from https://bpspsychub.onlinelibrary.wiley.com/doi/10.1111/bmsp.12327 by University Of North Florida, Wiley Online Library on [22/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

variance–covariance matrix of full rank. In this case, we write $\mathbf{Z}_j = [\mathbf{Z}_{1j}, \mathbf{Z}_{2j}]$ and, compatibly, $\boldsymbol{b}_j = [\boldsymbol{b}'_{1j}, \boldsymbol{b}'_{2j}]'$, where the $n_j \times q_1$ block $\mathbf{Z}_{1j}$ corresponds to the q_1 random effects in $\boldsymbol{b}_{1j}$ with a $q_1 \times q_1$ variance–covariance matrix $\boldsymbol{\Psi}_{11}$ of full rank; the $n_j \times q_2$ block $\mathbf{Z}_{2j}$ corresponds to the q_2 random effects in $\boldsymbol{b}_{2j}$ that are present in the model but not present in the population. In this way, the random-effect covariance matrix $\boldsymbol{\Psi}$ is also blocked as

$$\boldsymbol{\Psi} = \begin{bmatrix} \boldsymbol{\Psi}_{11} & \boldsymbol{\Psi}_{12} \\ \boldsymbol{\Psi}_{21} & \boldsymbol{\Psi}_{22} \end{bmatrix},$$

where $\boldsymbol{\Psi}_{22}$, $\boldsymbol{\Psi}_{21}$, and $\boldsymbol{\Psi}_{12}$ are zero matrices but as parameters they are still present in the model.

With this formulation, we have the following expressions that will be used frequently in our subsequent derivations:

$$\begin{aligned}
\mathbf{V}_j &= \mathbf{Z}_{1j}\boldsymbol{\Psi}_{11}\mathbf{Z}'_{1j} + \sigma^2 I_{n_j}, \\
\mathbf{V}_j^{-1} &= \frac{1}{\sigma^2}\left\{ I_{n_j} - \frac{1}{n_j}\mathbf{Z}_{1j}\mathbf{S}^{-1}_{Z_1Z_1,n_j}\mathbf{Z}'_{1j} \right\} + \frac{1}{n_j^2}\mathbf{Z}_{1j}\mathbf{S}^{-1}_{Z_1Z_1,n_j}\left[\frac{\sigma^2}{n_j}\mathbf{S}^{-1}_{Z_1Z_1,n_j} + \boldsymbol{\Psi}_{11}\right]^{-1}\mathbf{S}^{-1}_{Z_1Z_1,n_j}\mathbf{Z}'_{1j}, \\
\mathbf{V}_j^{-2} &= \frac{1}{\sigma^4}\left\{ I_{n_j} - \frac{1}{n_j}\mathbf{Z}_{1j}\mathbf{S}^{-1}_{Z_1Z_1,n_j}\mathbf{Z}'_{1j} \right\} \\
&\quad + \frac{1}{n_j^3}\mathbf{Z}_{1j}\mathbf{S}^{-1}_{Z_1Z_1,n_j}\left[\frac{\sigma^2}{n_j}\mathbf{S}^{-1}_{Z_1Z_1,n_j} + \boldsymbol{\Psi}_{11}\right]^{-1}\mathbf{S}^{-1}_{Z_1Z_1,n_j}\left[\frac{\sigma^2}{n_j}\mathbf{S}^{-1}_{Z_1Z_1,n_j} + \boldsymbol{\Psi}_{11}\right]^{-1}\mathbf{S}^{-1}_{Z_1Z_1,n_j}\mathbf{Z}'_{1j}.
\end{aligned}$$

B.2 | Information matrix of fixed effects (when $\boldsymbol{\Psi}$ is singular)

Below we re-evaluate the asymptotic behaviour of the information matrix $\mathbf{I}$. For the block $\mathbf{I}_{\boldsymbol{\beta\beta}}$ for fixed effects, similar to our derivations in Appendix A.1, we write

$$\mathbf{X}_j = \mathbf{Z}_{1j}\mathbf{W}_{1j} + \mathbf{E}_{1j}$$

for some $q_1 \times p$ matrix $\mathbf{W}_{1j}$ and some $n_j \times p$ matrix $\mathbf{E}_{1j}$, with $\mathbf{E}'_{1j}\mathbf{Z}_j = \mathbf{O}$ (a $p \times q_1$ zero matrix), and define $\mathbf{S}_{E_1E_1,n_j} = \mathbf{E}'_{1j}\mathbf{E}_{1j}/n_j$. Similar to Equation (A2), we obtain

$$\mathbf{I}_{\boldsymbol{\beta\beta}} = \frac{1}{\sigma^2}\sum_j n_j \mathbf{S}_{E_1E_1,n_j} + \sum_j \mathbf{W}'_{1j}\left[\frac{\sigma^2}{n_j}\mathbf{S}^{-1}_{Z_1Z_1,n_j} + \boldsymbol{\Psi}_{11}\right]^{-1}\mathbf{W}_{1j}. \tag{B1}$$

Defining $\overline{\mathbf{S}}_{E_1E_1} = \frac{1}{N}\sum_j n_j \mathbf{S}_{E_1E_1,n_j}$ and letting its limiting rank be $\tilde{p}_1$, we obtain

$$\log|\mathbf{I}_{\boldsymbol{\beta\beta}}| = \tilde{p}_2\log(J) + \tilde{p}_1\log(N) + O_p(1) = p\log(J) + \tilde{p}_1\log(\overline{n}) + O_p(1), \tag{B2}$$

where $\tilde{p}_2 = p - \tilde{p}_1$.

B.3 | Information matrix of random-effect parameters (when $\boldsymbol{\Psi}$ is singular)

For the block $\mathbf{I}_{\boldsymbol{\rho\rho}}$ corresponding to the random-effect parameters, we first write

$$\mathbf{Z}_{2j} = \mathbf{Z}_{1j}\overline{\mathbf{W}}_j + \overline{\mathbf{E}}_j$$

for some $q_1 \times q_2$ matrix $\overline{\mathbf{W}}_j$ and some $n_j \times q_2$ matrix $\overline{\mathbf{E}}_j$, with $\overline{\mathbf{E}}_j'\mathbf{Z}_{1,j} = \mathbf{O}$ (a $q_2 \times q_1$ zero matrix). Because $\mathbf{Z}_j$ has full column rank, the matrix $\mathbf{S}_{\overline{\mathbf{EE}},n_j} = \overline{\mathbf{E}}_j'\overline{\mathbf{E}}_j/n_j$ also has full rank.

The block $\mathbf{I}_{\boldsymbol{\rho\rho}}$ of parameters in the random effects is more complicated because blocks $\boldsymbol{\Psi}_{22}$ and $\boldsymbol{\Psi}_{21}(=\boldsymbol{\Psi}_{12}')$ need to be treated separately. We now define $\boldsymbol{\psi}_{rr} = \text{vech}\boldsymbol{\Psi}_{rr}$ $(r = 1, 2)$ and $\boldsymbol{\psi}_{12} = \text{vec}\boldsymbol{\Psi}_{12}$. The blocks of $\mathbf{I}_j$ are given by

$$
\begin{aligned}
\mathbf{I}_{\boldsymbol{\psi}_{rr}\boldsymbol{\psi}_{ss},j} &= \frac{1}{2}\mathbf{D}_{q_r}'\left\{\left[\mathbf{Z}_{r,j}'\mathbf{V}_j^{-1}\mathbf{Z}_{s,j}\right]\otimes\left[\mathbf{Z}_{r,j}'\mathbf{V}_j^{-1}\mathbf{Z}_{s,j}\right]\right\}\mathbf{D}_{q_s} \qquad (r,s=1,2),\\
\mathbf{I}_{\boldsymbol{\psi}_{12}\boldsymbol{\psi}_{ss},j} &= \frac{1}{2}\left\{\left[\mathbf{Z}_{2,j}'\mathbf{V}_j^{-1}\mathbf{Z}_{s,j}\right]\otimes\left[\mathbf{Z}_{1,j}'\mathbf{V}_j^{-1}\mathbf{Z}_{s,j}\right]+\left[\mathbf{Z}_{1,j}'\mathbf{V}_j^{-1}\mathbf{Z}_{s,j}\right]\otimes\left[\mathbf{Z}_{2,j}'\mathbf{V}_j^{-1}\mathbf{Z}_{s,j}\right]\right\}\mathbf{D}_{q_s} \qquad (s=1,2),\\
\mathbf{I}_{\boldsymbol{\psi}_{12}\boldsymbol{\psi}_{12},j} &= \frac{1}{2}\Big(\left[\mathbf{Z}_{1,j}'\mathbf{V}_j^{-1}\mathbf{Z}_{1,j}\right]\otimes\left[\mathbf{Z}_{2,j}'\mathbf{V}_j^{-1}\mathbf{Z}_{2,j}\right]+\left[\mathbf{Z}_{1,j}'\mathbf{V}_j^{-1}\mathbf{Z}_{2,j}\right]\otimes\left[\mathbf{Z}_{2,j}'\mathbf{V}_j^{-1}\mathbf{Z}_{1,j}\right]\\
&\quad+\left[\mathbf{Z}_{2,j}'\mathbf{V}_j^{-1}\mathbf{Z}_{1,j}\right]\otimes\left[\mathbf{Z}_{1,j}'\mathbf{V}_j^{-1}\mathbf{Z}_{2,j}\right]+\left[\mathbf{Z}_{2,j}'\mathbf{V}_j^{-1}\mathbf{Z}_{2,j}\right]\otimes\left[\mathbf{Z}_{1,j}'\mathbf{V}_j^{-1}\mathbf{Z}_{1,j}\right]\Big),\\
\mathbf{I}_{\boldsymbol{\psi}_{rr}\sigma^2,j} &= \frac{1}{2}\mathbf{D}_{q_r}'\text{vec}(\mathbf{Z}_{r,j}'\mathbf{V}^{-2}\mathbf{Z}_{r,j}) \qquad (r=1,2) \quad \text{and}\\
\mathbf{I}_{\boldsymbol{\psi}_{12}\sigma^2,j} &= \frac{1}{2}\text{vec}(\mathbf{Z}_{2,j}'\mathbf{V}^{-2}\mathbf{Z}_{1,j}+\mathbf{Z}_{1,j}'\mathbf{V}^{-2}\mathbf{Z}_{2,j}),\\
\mathbf{I}_{\sigma^2\sigma^2,j} &= \frac{n_j-q_1}{2\sigma^4}+\frac{1}{2n_j^2}\text{tr}\left\{\left[\frac{\sigma^2}{n_j}\mathbf{I}_{q_1}+\boldsymbol{\Psi}_{11}\mathbf{S}_{Z_1Z_1,n_j}\right]^{-2}\right\},
\end{aligned}
$$

where

$$
\begin{aligned}
\mathbf{Z}_{1,j}'\mathbf{V}_j^{-1}\mathbf{Z}_{1,j} &= \left[\boldsymbol{\Psi}_{11}+\frac{\sigma^2}{n_j}\mathbf{S}_{Z_1Z_1,n_j}^{-1}\right]^{-1},\\
\mathbf{Z}_{2,j}'\mathbf{V}_j^{-1}\mathbf{Z}_{2,j} &= \frac{n_j}{\sigma^2}\mathbf{S}_{\overline{\mathbf{EE}},n_j}+\overline{\mathbf{W}}_j'\left[\frac{\sigma^2}{n_j}\mathbf{S}_{Z_1Z_1,n_j}^{-1}+\boldsymbol{\Psi}_{11}\right]^{-1}\overline{\mathbf{W}}_j,\\
\mathbf{Z}_{2,j}'\mathbf{V}_j^{-1}\mathbf{Z}_{1,j} &= \overline{\mathbf{W}}_j'\left[\frac{\sigma^2}{n_j}\mathbf{S}_{Z_1Z_1,n_j}^{-1}+\boldsymbol{\Psi}_{11}\right]^{-1},\\
\mathbf{Z}_{1,j}'\mathbf{V}_j^{-2}\mathbf{Z}_{1,j} &= \frac{1}{n_j}\left[\frac{\sigma^2}{n_j}\mathbf{S}_{Z_1Z_1,n_j}^{-1}+\boldsymbol{\Psi}_{11}\right]^{-1}\mathbf{S}_{Z_1Z_1,n_j}^{-1}\left[\frac{\sigma^2}{n_j}\mathbf{S}_{Z_1Z_1,n_j}^{-1}+\boldsymbol{\Psi}_{11}\right]^{-1},\\
\mathbf{Z}_{2,j}\mathbf{V}_j^{-2}\mathbf{Z}_{2,j} &= \frac{n_j}{\sigma^4}\mathbf{S}_{\overline{\mathbf{EE}},n_j}+\frac{1}{n_j}\overline{\mathbf{W}}'\left[\frac{\sigma^2}{n_j}\mathbf{S}_{Z_1Z_1,n_j}^{-1}+\boldsymbol{\Psi}_{11}\right]^{-1}\mathbf{S}_{Z_1Z_1,n_j}^{-1}\left[\frac{\sigma^2}{n_j}\mathbf{S}_{Z_1Z_1,n_j}^{-1}+\boldsymbol{\Psi}_{11}\right]^{-1}\overline{\mathbf{W}},\\
\mathbf{Z}_{2,j}\mathbf{V}_j^{-2}\mathbf{Z}_{1,j} &= \frac{1}{n_j}\overline{\mathbf{W}}'\left[\frac{\sigma^2}{n_j}\mathbf{S}_{Z_1Z_1,n_j}^{-1}+\boldsymbol{\Psi}_{11}\right]^{-1}\mathbf{S}_{Z_1Z_1,n_j}^{-1}\left[\frac{\sigma^2}{n_j}\mathbf{S}_{Z_1Z_1,n_j}^{-1}+\boldsymbol{\Psi}_{11}\right]^{-1}.
\end{aligned}
$$

From these results we can see the information matrix of the random-effect parameters has the following order (the information matrix is symmetric, with only the lower-diagonal part filled in):

$$
\mathbf{I}_{\boldsymbol{\rho\rho},j} = \begin{bmatrix}
\mathbf{I}_{\boldsymbol{\psi}_{11}\boldsymbol{\psi}_{11},j} & & & \\
\mathbf{I}_{\boldsymbol{\psi}_{12}\boldsymbol{\psi}_{11},j} & \mathbf{I}_{\boldsymbol{\psi}_{12}\boldsymbol{\psi}_{12},j} & & \\
\mathbf{I}_{\boldsymbol{\psi}_{22}\boldsymbol{\psi}_{11},j} & \mathbf{I}_{\boldsymbol{\psi}_{22}\boldsymbol{\psi}_{12},j} & \mathbf{I}_{\boldsymbol{\psi}_{22}\boldsymbol{\psi}_{22},j} & \\
\mathbf{I}_{\sigma^2\boldsymbol{\psi}_{11},j} & \mathbf{I}_{\sigma^2\boldsymbol{\psi}_{12},j} & \mathbf{I}_{\sigma^2\boldsymbol{\psi}_{22},j} & \mathbf{I}_{\sigma^2\sigma^2,j}
\end{bmatrix} = \begin{bmatrix}
O_p(1) & & & \\
O_p(1) & O_p(n_j) & & \\
O_p(1) & O_p(n_j) & O_p(n_j^2) & \\
o_p(1) & o_p(1) & O_p(n_j) & O_p(n_j)
\end{bmatrix}.
$$

20448317, 2024, 2, Downloaded from https://bpspsychub.onlinelibrary.wiley.com/doi/10.1111/bmsp.12327 by University Of North Florida, Wiley Online Library on [22/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Summing over the blocks $j = 1, 2, \ldots, J$, for the case of equal cluster sizes where $n_j = n$, as $n, J \to \infty$, the log-determinant is

$$\begin{aligned}\log|\mathbf{I}_{\boldsymbol{\rho}\boldsymbol{\rho}}| &= (q_1 q_2 + 1)\log(nJ) + q_2^* \log(n^2 J) + q_1^* \log(J) + O_p(1) \\ &= (q^* + 1)\log(J) + (2q_2^* + q_1 q_2 + 1)\log(n) + O_p(1),\end{aligned}$$

where $q^* = q(q+1)/2$ and $q_r^* = q_r(q_r+1)/2$ for $r = 1, 2$.

20448317, 2024, 2, Downloaded from https://bpspsychub.onlinelibrary.wiley.com/doi/10.1111/bmsp.12327 by University Of North Florida, Wiley Online Library on [22/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License